DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

PROJECT -I REPORT

Submitted by

A.KANNAN	**(Reg. No: 95192309015)**
R.MUTHU BALA	**(Reg. No: 95192309028)**
M.SANTHOSH	**(Reg. No: 95192309044)**
K.AKASH	**(Reg. No: 95192309701)**

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ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this Project - I Report **"DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES"** is the bonafide work of **A.KANNAN (95192309015), R.MUTHU BALA (95192309028), M.SANTHOSH (95192309044) and K.AKASH (95192309701),** who carried out the project work under my supervision.

SIGNATURE
Mrs. V. DEVINANDHINI, M.E.,
SUPERVISOR,
Assistant Professor,
Department of AI & DS,
P.S.R. Engineering College
Sivakasi – 626140

SIGNATURE
Dr. C. NATARAJAN, M.E., Ph.D.,
HEAD OF THE DEPARTMENT,
Associate Professor,
Department of AI & DS,
P.S.R. Engineering College
Sivakasi – 626140

Submitted for the PROJECT - I Viva-Voce Examination held on ……

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

The Smart Attendance System is a contactless institutional attendance platform that automates student attendance using multi-face recognition and timetable-driven session management. It removes the need for manual roll calls, QR-code scanning, shared identity cards and direct-contact biometric devices, and provides separate Student, Staff and Administrator workflows for enrollment, live attendance, attendance history, timetable management, reporting and system administration, secured with JWT authentication and role-based access control. The biometric pipeline combines OpenCV YuNet multi-face detection, five-point facial alignment, face-quality assessment, 512-dimensional ArcFace/SFace-style deep embeddings, cosine-similarity matching, liveness analysis and multi-frame temporal verification, so that a single noisy frame can never become a final attendance event. Students are enrolled from multiple face angles, and the system records first-seen and last-seen timestamps against the academic timetable slot that is active at that moment. The platform is built on a Python FastAPI backend, a SQLite database accessed through SQLAlchemy, and a lightweight HTML, CSS and JavaScript interface, and it exports attendance in CSV, Excel and PDF formats. An Academic Evaluation module compares a classical Haar-cascade baseline, derived from the authors' conference paper, with the proposed deep pipeline using accuracy, precision, recall, F1-score, FAR, FRR, latency and FPS. Smart Attendance System demonstrates that attendance can be treated as an auditable academic process in which every record is traceable to a timetable session, a verified identity and an authorised user.

TABLE OF CONTENTS

S.NO	TITLE	PAGE NO.
	Abstract	iv
	List of Figures	viii
	List of Tables	ix
	List of Abbreviations	x
1	**INTRODUCTION**	1
	1.1 Background and Motivation	1
	1.2 Problem Statement	2
	1.3 System Overview	2
	1.4 Technology Stack	3
	1.5 Scope and Limitations	4
	1.6 Stakeholders	4
	1.7 Report Organization	5
2	**LITERATURE REVIEW**	6
	2.1 Review of Related Works	6
	2.2 Comparative Analysis of Reviewed Approaches	9
	2.3 Summary and Research Gap	10
3	**SYSTEM ANALYSIS**	13
	3.1 Problem Identification	13
	3.2 Objectives	13
	3.3 Functional Requirements	14
	3.4 Non-Functional Requirements	16
	3.5 System Modules	17
	3.6 Use Case Description	19
	3.7 Data Flow Diagram	20
	3.8 System Architecture	21
	3.9 Database Schema	22
	3.10 Feasibility Study	24
	3.11 Risk Analysis	25
	3.12 Assumptions and Dependencies	26
4	**SYSTEM DESIGN**	28
	4.1 Backend Architecture	28
	4.2 Frontend Architecture	28
	4.3 Security Design	29
	4.4 REST API Design	29

S.NO	TITLE	PAGE NO.
	4.5 Database ER Diagram Description	31
	4.6 AI Service Design	31
	4.7 Sequence Flow	33
	4.8 Error Handling and Resilience Design	35
	4.9 Module Interaction Design	36
	4.10 Deployment Architecture	36
5	**IMPLEMENTATION**	37
	5.1 Authentication and RBAC Module	37
	5.2 Student Management	38
	5.3 Biometric Enrollment	39
	5.4 Face Detection	39
	5.5 Face Alignment and Quality Assessment	40
	5.6 ArcFace Embedding	40
	5.7 Face Matching	41
	5.8 Liveness Detection	41
	5.9 Temporal Verification	42
	5.10 Timetable Scheduler	42
	5.11 Live Attendance	42
	5.12 Student Portal	43
	5.13 Staff and Admin Dashboards	44
	5.14 Reporting Module	45
	5.15 Academic Evaluation Module	45
	5.16 Audit and Configuration	46
6	**TESTING**	47
	6.1 Testing Strategy	47
	6.2 Functional Test Cases	48
	6.3 Biometric Pipeline Testing	50
	6.4 Role and Authorization Testing	51
	6.5 Timetable and Session Testing	51
	6.6 Reporting and Export Testing	51
	6.7 Security and Isolation Testing	51
	6.8 Error Handling Testing	52
	6.9 Requirements Traceability	52
	6.10 Manual Verification Walkthrough	53

S.NO	TITLE	PAGE NO.
		55
7	**RESULTS AND DISCUSSION**	
	7.1 System Performance Results	55
	7.2 Functional Outcomes	57
	7.3 Existing vs Proposed System	58
	7.4 Discussion	60
	7.5 Positioning Against Generic Alternatives	61
	7.6 Threats to Validity	62
8	**CONCLUSION AND FUTURE WORK**	63
	8.1 Conclusion	63
	8.2 Future Enhancement	63
	REFERENCES	65
	APPENDIX	68

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	System Context / Data Flow Diagram — Level 0	21
3.2	Smart Attendance System Architecture	22
3.3	Entity-Relationship Diagram	24
4.1	Role and Security Flow	29
4.2	Proposed Face Recognition Pipeline	32
4.3	Sequence Flow of a Live Attendance Request	34
4.4	Timetable-Driven Attendance Session Flow	35
5.1	Login and Role Selection	37
5.2	Student Management and Enrollment	38
5.3	Multi-Angle Biometric Enrollment	39
5.4	Face Detection and Alignment	40
5.5	Live Attendance State Workflow	43
5.6	Student Dashboard	44
5.7	Staff and Administrator Dashboard	44
5.8	Report Generation Workflow	45
5.9	Academic Evaluation Workflow	46
5.10	Audit Log and System Settings	46
6.1	Functional Testing Flow	47
6.2	Biometric Test Conditions	50
7.1	Published Baseline Recognition Accuracy by Cohort	56
7.2	Attendance Outcome Flow	58
7.3	Baseline vs Proposed Pipeline Comparison	60

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
1.1	Smart Attendance System Technology Stack	3
1.2	Project Stakeholders	5
2.1	Comparative Summary of Automated Attendance Literature	9
2.2	Baseline Research Pipeline and Implemented Pipeline	10
2.3	Reviewed Sources Mapped to Smart Attendance Design Decisions	11
3.1	Functional Requirements	16
3.2	Non-Functional Requirements	17
3.3	System Module Summary	18
3.4	Use Case Descriptions	20
3.5	Database Schema — Table Definitions	23
3.6	Risk Analysis	26
3.7	Assumptions and Their Consequences	27
4.1	Representative REST API Endpoint Groups	31
4.2	Vision Pipeline Stages	33
4.3	Error Cases and Designed Responses	36
5.1	Role Permissions	38
5.2	Live Attendance States	43
6.1	Test Suite Summary	48
6.2	Functional Test Cases	50
6.3	Biometric Test Conditions and Expected Behaviour	51
6.4	Security and Authorization Tests	52
6.5	Requirements Traceability Matrix	53
7.1	Evaluation Metrics and Their Purpose	55
7.2	Published Baseline Performance Reported in the Conference Paper [1]	56
7.3	Functional Outcomes Against Objectives	58
7.4	Existing Systems vs Proposed System — Comparative Analysis	59
7.5	Smart Attendance System vs Generic Alternatives	62

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
ArcFace	Additive Angular Margin Face Recognition Loss / Embedding
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
DFD	Data Flow Diagram
EAR	Eye Aspect Ratio
ER	Entity-Relationship
FAR	False Acceptance Rate
FPS	Frames Per Second
FQA	Face Quality Assessment
FRR	False Rejection Rate
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
JWT	JSON Web Token
KNN	K-Nearest Neighbors
LBPH	Local Binary Pattern Histograms
MFIAS	Multi-Facet Institutional Attendance System
ONNX	Open Neural Network Exchange
ORM	Object-Relational Mapping
PCA	Principal Component Analysis
PDF	Portable Document Format
RBAC	Role-Based Access Control
REST	Representational State Transfer
SFace	Sigmoid-Constrained Hypersphere Face Recognition Model
SQL	Structured Query Language
UI	User Interface
XLSX	Office Open XML Spreadsheet Format
YOLO	You Only Look Once
YuNet	Lightweight Millisecond-level Face Detector

CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

Attendance is one of the most basic administrative requirements of an educational institution. Attendance records support internal assessment eligibility, laboratory and examination approval, scholarship and accreditation reporting, and early identification of students who are drifting away from their studies. Despite the many technological advances of recent years, attendance in most classrooms is still taken in an old-fashioned way: a faculty member calls out names, a sign-in sheet is passed around, or a student is asked to carry an identity card or tap a fingerprint scanner. The authors' conference paper [1] reports that manual roll calls alone can consume a substantial fraction of a lecture period, that paper sheets remain open to buddy signing, and that fingerprint devices are unhygienic and slow when a large group of students must use them in a short interval.

Mobile applications and QR-code scanning were introduced to reduce this burden, but they replace one weakness with another. A QR code displayed on a projector can be photographed and forwarded to a friend outside the room, and every student must still find the application, sign in and point a phone at the screen, which interrupts teaching. Contactless computer vision offers a practical alternative. A camera can observe several students at once, detect their faces and compare them with enrolled biometric representations without any action on the student's part, and because the face is physically present at the point of capture it is far harder to share than a code or a card.

Ready-made face recognition, however, performs noticeably worse in unstructured classrooms than in controlled laboratory conditions. Lighting varies between bright daylight and dim interiors, students do not look straight at the camera, spectacles, beards and caps hide geometric landmarks, and an overly heavy pipeline delays every entry. These concerns, identified in the conference paper [1] and in the related literature reviewed in Chapter 2, motivated the present project: a complete institutional attendance platform that does not stop at recognising a face but also decides *when* attendance may be taken, *who* may change a record, and *how* the evidence is stored and reported.

1.2 Problem Statement

There is a need for an attendance system that works without manual roll call, QR scanning or physical contact while still keeping institutional control over student records. The system must distinguish enrolled students from unknown faces, handle several faces in a single camera frame, record attendance with timestamps against the correct class period, and give authorised staff reliable reports. Existing lightweight pipelines are frequently evaluated against one disturbance at a time, whereas a real classroom combines lighting variation, head-pose changes and accessories simultaneously, and a recognition result that is not tied to an active timetable session or an accountable user is of little administrative value.

The implementation must also reduce false attendance caused by poor-quality frames, momentary detections and presentation attacks such as holding up a photograph. The project therefore combines face-quality assessment, multi-frame temporal verification and liveness analysis rather than treating a single frame as sufficient evidence, and it surrounds the recognition core with timetable-driven session control, role-based access, audit logging and multi-format reporting.

1.3 System Overview

The Smart Attendance System is a web-based application with a Python FastAPI backend, a browser-based frontend and SQLite persistence. The backend exposes REST interfaces for authentication, student management, biometric enrollment, timetable management, attendance sessions, live frame processing, dashboards, reports and academic evaluation. The frontend is a single-page interface whose navigation is organised by role, so that a student, a staff member and an administrator each see only the functions that belong to them.

The biometric path is implemented as a sequence of independent stages: YuNet face detection, five-point alignment, face-quality assessment, deep face embedding, cosine-similarity matching, liveness checking and temporal confirmation. A timetable scheduler maps the current time to an academic slot and creates or reuses the corresponding attendance session, so the camera pipeline is always interpreted inside the correct class period. Each confirmed student is stored with first-seen and last-seen timestamps, giving the institution evidence of when a student entered and when the student was last observed.

The system deliberately separates the biometric models from the application logic. The detector and recogniser are stored as ONNX assets and are used only for inference, so no neural network is trained while attendance is being taken; the enrolled gallery is simply a set of stored embeddings that can be extended whenever a new student is registered.

1.4 Technology Stack

Table 1.1 summarises the technology stack used across the application, the biometric pipeline and the reporting layer. Python is used throughout the server side because the computer-vision, numerical and reporting libraries on which the project depends are all first-class Python packages, and FastAPI provides typed request validation and automatic interface documentation with very little boilerplate.

Layer	Technology	Role
Frontend	HTML, CSS, JavaScript	Role-specific dashboards, enrollment forms and camera workflow
Backend	Python, FastAPI, Uvicorn, Pydantic	REST APIs, request validation and application logic
Computer Vision	OpenCV, NumPy, YuNet, SFace/ArcFace (ONNX)	Detection, alignment, embedding and similarity matching
Liveness and Tracking	Eye aspect ratio, micro-motion, temporal tracker	Presentation-attack evidence and multi-frame stability
Database	SQLite, SQLAlchemy ORM	Students, embeddings, timetable, sessions, attendance, audit and configuration data
Security	JWT, role checks, password hashing	Authentication, authorisation and protected routes
Reports	CSV, OpenPyXL, ReportLab	Institutional report export in CSV, Excel and PDF formats
Scheduling	Background timetable scheduler	Automatic creation of attendance sessions from timetable slots

Table 1.1: Smart Attendance System Technology Stack

1.5 Scope and Limitations

The scope of the project covers role-based authentication and authorisation; student registration and management; multi-angle biometric enrollment; multi-face detection, alignment and face-quality checks; deep embedding generation and similarity matching; liveness analysis and temporal verification; timetable-driven attendance sessions; live attendance with first-seen and last-seen timestamps; student, staff and administrator dashboards; audit logging and configurable thresholds; and CSV, Excel and PDF reporting. An Academic Evaluation module is included so that the baseline and proposed pipelines can be compared on the institution's own enrolled data.

The limitations are stated openly rather than left implicit. Recognition quality depends on camera placement, lighting and the quality of the enrolled samples, and the recognition and liveness thresholds are configurable values that must be calibrated for each deployment rather than copied from another environment. This report does not present a final benchmark score that was not produced by an executed evaluation: the accuracy and latency figures quoted from the authors' conference paper [1] are treated as published baseline evidence and not as a measurement of this implementation. The liveness module is designed to resist simple static presentations; it is not claimed to defeat sophisticated attacks such as replayed video or three-dimensional masks, which remain future work.

1.6 Stakeholders

The stakeholders considered during requirements gathering and design, and the interest each has in the behaviour of the platform, are identified in Table 1.2.

Stakeholder	Primary interest
Student	A contactless, fast attendance experience; visibility of personal attendance history and percentage; confidence that records are accurate and cannot be changed by other students
Staff / Faculty	Reliable live attendance for the active class, simple correction-free reports, and read access to student records and session history
Administrator	Control over student enrollment, timetable, subjects, faculty, thresholds and corrections, with an audit trail of every administrative change

Stakeholder	Primary interest
Institution / Management	Auditable attendance data for eligibility, accreditation and compliance, produced without extra clerical effort
Academic reviewers	Evidence that the design decisions are traceable to the literature and that performance claims are separated from measurements

Table 1.2: Project Stakeholders

1.7 Report Organization

The remainder of this report is organised as follows. Chapter 2 reviews the literature on automated attendance, classical and deep face recognition and presentation-attack detection, and identifies the research gap addressed by the project. Chapter 3 analyses the system through its objectives, functional and non-functional requirements, modules, use cases, data flow, architecture, database schema, feasibility, risks and assumptions. Chapter 4 describes the design of the backend, frontend, security layer, REST interfaces, biometric service, sequence flow, error handling and deployment. Chapter 5 documents the implementation of each module. Chapter 6 describes the testing strategy, test cases and traceability, Chapter 7 discusses results and compares the proposed system with existing approaches, and Chapter 8 concludes the report and outlines future enhancements. References and appendices follow.

CHAPTER 2
LITERATURE REVIEW

2.1 Review of Related Works

Automated attendance has been studied for more than two decades. Early systems were built on the Viola-Jones object detector [13], which evaluates Haar-like rectangular features on an integral image and rejects non-face regions through a cascade of boosted classifiers, and they usually paired it with Local Binary Pattern Histograms or Principal Component Analysis for identification. Such pipelines need little computation and can run on very modest hardware, but, as the authors' conference paper [1] observes, they degrade in extreme lighting, with small head movements and with partially hidden faces.

Bairagi, R., Ahmed, R., Tisha, S. A. and co-authors (2021) [3], in a paper presented at the International Conference on Inventive Research in Computing Applications, describe a real-time attendance system built on a Haar classifier. The approach is attractive for its minimal cost and fast record writing, but it fails under off-pose faces and lighting shifts. The present project retains a Haar-cascade pipeline only as the baseline inside its Academic Evaluation module, so that the benefit of a modern detector can be measured instead of assumed.

Trivedi, A., Tripathi, C. M., Perwej, Y. and co-authors (2022) [4], in the International Journal of Scientific Research in Science and Technology, present a face recognition attendance system based on OpenCV and Eigenfaces that automates attendance logging into a SQL database. The system shows how little infrastructure a working prototype needs, but its alignment is rigid and it has no tolerance for occlusion. Its database-centred design nevertheless supports the decision in this project to keep every attendance record in a relational store with clear references to the student and the session.

Pattnaik, P. and Mohanty, K. K. (2020) [9] compare artificial-intelligence techniques for real-time face-recognition-based attendance, including cloud vision services such as Google Cloud Vision and Amazon Rekognition. Cloud services offer high accuracy on still images, but when applied to continuous classroom video they produce heavy network traffic and unpredictable latency, and they require a stable connection from the institution's cameras to an external provider, which also raises a privacy concern. This review is one of the reasons

the proposed system performs all inference locally on pretrained ONNX models instead of sending video to a remote service.

Lateef, A. S. and Kamil, M. Y. (2023) [6] use several YOLOv7 models to track many faces simultaneously in a smart classroom. Continuous multi-face tracking is the correct goal for an attendance system, but running such detectors at real-time frame rates requires expensive GPU hardware, with high latency at edge nodes, which is impractical for most institutions. Febri Ananda, G., Nugroho, H. A. and Ardiyanto, I. (2025) [8] address a related classroom problem, the recognition of low-resolution faces, using YOLOv8, which confirms that distance from the camera and image resolution are real obstacles in lecture halls. The proposed system therefore chooses a lightweight detector, YuNet [19], designed for millisecond-level inference on ordinary CPUs, and adds a face-quality gate so that unusable small or blurred faces are deferred instead of being forced into a poor decision.

Kamil, M. H. M., Zaini, N., Mazalan, L. and Ahamad, A. H. (2023) [5], in Multimedia Tools and Applications, propose an attendance system with mask recognition that combines a support vector machine with a mask-oriented CNN and reaches 81.8% accuracy on masked faces. The work shows that occlusion can be handled when it is the only disturbance, but it addresses isolated occlusions and does not handle pose variation. Afiyanto, R., Astuti, E. D., Kamal, A. A. and Santoso, N. P. L. (2025) [7] assess attendance systems under different lighting settings and report that distance-based matching degrades by 34% in classrooms illuminated at 150 lux or less. Together these studies justify two features of the project: multi-angle enrollment, which stores several views of each student, and a face-quality assessment stage that rejects samples whose illumination or sharpness is inadequate.

Jack, K. E. and co-authors (2026) [2], published in Discover Artificial Intelligence, develop a multi-facet facial recognition model for institutional attendance activities. Their decoupled hybrid pipeline combines Haar-cascade detection, Dlib landmark alignment, a ResNet-34 128-dimensional vector encoding and a distance-weighted K-nearest-neighbour classifier, and is reported to reach 99.29% recognition accuracy on video streams with a 6.00-second average verification time. The authors' own conference paper [1] builds on this model into the Multi-Facet Institutional Attendance System (MFIAS), which adds an automatic departmental timetable engine, simultaneous multi-student verification, entry and exit timestamps, administrator-only record changes and CSV, Excel and PDF reporting. The present project takes MFIAS as its research foundation and baseline.

The deep-embedding techniques on which modern recognition rests are well established. He, K., Zhang, X., Ren, S. and Sun, J. (2016) [15] introduced residual learning, which allows very deep networks such as ResNet-34 to be trained without vanishing gradients. Schroff, F., Kalenichenko, D. and Philbin, J. (2015) [16] showed with FaceNet that a network can be trained with a triplet loss to map faces into a compact Euclidean space in which distance directly measures identity similarity. King, D. E. (2009) [14] released Dlib, whose 68-point landmark model and 128-dimensional face encoder were used in the baseline pipeline. Deng, J., Guo, J., Xue, N. and Zafeiriou, S. (2019) [17] proposed ArcFace, an additive angular margin loss that produces highly discriminative embeddings compared by angle, and Zhong, Y. and Deng, W. (2021) [18] proposed SFace, a sigmoid-constrained hypersphere loss that improves the robustness of the learned embedding and is available as a compact model through OpenCV. These angular-margin embeddings are compared by cosine similarity, which is the matching rule adopted in this project.

Presentation attacks are a recognised weakness of any camera-based biometric. Soukupova, T. and Cech, J. (2016) [20] describe a real-time blink detector based on the eye aspect ratio computed from facial landmarks, a measure that is cheap enough to evaluate on every frame. The conference paper [1] itself lists infrared depth sensing and blink classifiers as future work for hardware-level anti-spoofing. This project implements a software-only liveness layer based on eye-aspect-ratio behaviour and micro-motion evidence, and keeps the result with the attendance record so that a successful match is always linked to the verification state that produced it.

Finally, several further studies treat attendance as a complete administrative workflow rather than a recognition benchmark. Patel, J., Suthar, S., Rathod, V. and Chavda, J. (2023) [10] study real-time attendance management with facial recognition and computer vision, Varute, M. T., Rajput, M. S., Shinde, M. S. and Shinde, M. P. (2024) [11] present a smart strategy that goes beyond the roll call, and Matar, N. A. (2025) [12] investigates a smart attendance system using face recognition in a master's thesis. Their continued appearance in the literature confirms that institutions still need integrated solutions, and their emphasis on workflow supports the project's decision to combine recognition with sessions, roles and reports.

2.2 Comparative Analysis of Reviewed Approaches

Reading the reviewed sources together shows a clear pattern: each addresses one slice of the attendance problem, and each slice has its own evaluation tradition. Table 2.1 summarises the principal approaches, their core methodology, their main advantage and the limitation or gap that remains. Classical Haar and LBPH systems [3] are evaluated on how cheaply they can be built; mask and illumination studies [5], [7] are evaluated on accuracy under a single disturbance; tracking studies [6] are evaluated on how many faces can be followed; and the MFIAS model [1], [2] is evaluated on recognition accuracy and verification time.

Reference	Core Methodology	Primary Advantage	Identified Limitation / Gap
Bairagi et al. (2021) [3]	Haar + LBPH	Minimal cost; fast record writing	Fails under off-pose and lighting shifts
Trivedi et al. (2022) [4]	OpenCV + Eigenfaces	Automates SQL logging	Rigid alignment; no occlusion tolerance
Kamil et al. (2023) [5]	SVM + mask CNN	Masked-face verification (81.8%)	Isolated occlusions only; no pose handling
Lateef and Kamil (2023) [6]	YOLOv7 deep model	Continuous multi-face tracking	Heavy GPU load; high latency at edge nodes
Afiyanto et al. (2025) [7]	Illumination study with CNN	Quantifies lighting sensitivity	High overhead; limited on composite angles
Jack et al. (2026) [2]	Haar + Dlib + ResNet-34 + KNN	Multi-facet robustness (99.29%)	6.00 s latency; lacks timetable automation
Conference paper MFIAS [1]	Timetable engine + ResNet + KNN	No QR; entry / exit logs	Needs calibrated entry camera placement; batching

Table 2.1: Comparative Summary of Automated Attendance Literature

This pattern means that no reviewed source measures an attendance system end to end. A system that detects faces well may still be useless if it marks attendance outside the class period, if an unknown person is accepted, or if a record can be changed without leaving a

trace. The review also exposes a difference in the unit of evaluation: published figures describe recognition of frames or images, whereas an institution cares about whether the right student is marked present in the right session once, with evidence.

The comparison also separates what the authors' baseline offers from what the implemented project adds. The baseline [1] uses Haar detection, 128-dimensional ResNet-style encodings, Euclidean distance with a distance-weighted KNN vote, and a timetable engine described in the paper; the implemented project uses YuNet detection with a fallback detector, five-point alignment and face-quality assessment, 512-dimensional ArcFace/SFace-style embeddings, cosine similarity with a configurable threshold, liveness verification and temporal tracking, together with an implemented timetable scheduler, role-based access and CSV, Excel and PDF reporting. Table 2.2 makes this correspondence explicit.

Dimension	**Baseline (conference paper)**	**Implemented project**
Detection	Haar Cascade	YuNet with fallback detector
Alignment	68-point landmark alignment	Five-point similarity alignment
Representation	128-D ResNet-34 encoding	512-D ArcFace/SFace-style embedding
Matching	Distance-weighted KNN (Euclidean)	Cosine similarity, configurable threshold
Robustness	Lighting, pose and accessories discussed	Multi-angle enrollment, FQA, liveness, temporal verification
Attendance control	Timetable engine in design	Implemented timetable scheduler and sessions
Reporting	CSV / Excel / PDF described	CSV / Excel / PDF implemented

Table 2.2: Baseline Research Pipeline and Implemented Pipeline

2.3 Summary and Research Gap

Consolidating the reviewed literature, the sources span six distinct concerns: classical face detection and its limits, robustness to occlusion and illumination, the cost of real-time multi-face tracking, deep metric embeddings, presentation-attack resistance, and timetable and governance features of an attendance product. Each concern is addressed by at least one

source, but none of the sources combines all of them in a single deployable workflow. Table 2.3 maps each concern to the reviewed sources and to the design decision it informed in this project.

Concern	Key sources	Design decision informed
Classical detection limits	[3], [4], [13]	Haar pipeline kept only as the evaluation baseline; YuNet adopted as the primary detector
Occlusion, pose and illumination	[1], [5], [7]	Multi-angle enrollment and a face-quality gate before matching
Cost of real-time multi-face tracking	[6], [8], [9]	Lightweight ONNX inference on ordinary CPUs; local processing without cloud video
Deep metric embeddings	[14] – [18]	512-D ArcFace/SFace-style embedding compared by cosine similarity
Presentation attacks	[1], [20]	Eye-aspect-ratio and micro-motion liveness evidence stored with each record
QR vulnerability and manual effort	[1], [2]	Timetable scheduler that opens sessions automatically
Governance and auditability	[1], [21]	JWT role-based access, configuration values and an audit log of administrative actions

Table 2.3: Reviewed Sources Mapped to Smart Attendance Design Decisions

The research gap adopted by this project is therefore broader than recognition accuracy. Attendance automation is not only a face-recognition problem: a usable institutional system must control when attendance is active, who may modify records, how unknown faces are handled, how evidence is logged and how administrators obtain reports. The project closes this gap by integrating biometric identification with timetable control, liveness analysis, temporal verification, authorisation and reporting in one deployable workflow, and by separating published research measurements from the measurements the implementation can produce on its own data.

Two concrete decisions follow from the review. First, a recognition result is never an attendance record by itself; it becomes one only when it is observed repeatedly and with liveness evidence inside an active timetable session. Second, the evaluation approach is built into the product through the Academic Evaluation module, so that the claimed advantage of the proposed pipeline over the Haar baseline can be reproduced on local data in the manner discussed in Chapters 6 and 7, rather than inherited from a paper written under different conditions.

CHAPTER 3
SYSTEM ANALYSIS

3.1 Problem Identification

Several interconnected problems motivate this system. First, manual attendance effort: calling names, collecting signatures and transcribing sheets into a register consumes teaching time and clerical effort, and the resulting records are only as reliable as the person who keeps them. Second, identity impersonation: a signature, a shared QR code or an identity card does not prove that the person who presented it is the student concerned, which is why proxy attendance persists in every manual or token-based scheme.

Third, recognition that is unstable in real conditions. A face recogniser that works on a clean frontal photograph can fail in a classroom where lighting changes, heads are turned, faces are partially hidden and several people stand in the frame at once. Fourth, a lack of timetable awareness: a recogniser that does not know which class is in progress cannot decide whether an observation belongs to a session or is a student merely walking past. Fifth, weak auditability: if records can be edited without an actor, a reason and a timestamp, the attendance register cannot be defended during a dispute or an audit.

A system that solves only the recognition step can therefore still produce incorrect attendance if a class is not active, if an unknown face is accepted, or if a record is changed without authorisation. The problem to be solved is consequently an end-to-end one, and the requirements that follow cover the whole chain from enrollment to reporting.

3.2 Objectives

The objectives of the Smart Attendance System, derived from the problems identified above, are as follows.

1. Automate attendance using contactless multi-face recognition, so that no manual roll call, QR scan or physical contact is required.
2. Enroll each student using multiple face angles to improve the robustness of the stored biometric representation.
3. Detect and reject poor-quality or suspicious frames before attendance is marked.
4. Use deep face embeddings and similarity matching for identity verification, with an explicit and configurable decision threshold.

5. Activate attendance sessions automatically according to the academic timetable.
6. Maintain first-seen and last-seen attendance timestamps for every student in every session.
7. Provide student, staff and administrator workflows with role-based access control.
8. Generate institutional attendance reports in CSV, Excel and PDF formats.
9. Provide an Academic Evaluation module that makes the comparison between a baseline pipeline and the proposed pipeline measurable and repeatable.

3.3 Functional Requirements

The functional requirements are listed in Table 3.1. Priorities are High for requirements without which the core attendance workflow cannot operate and Medium for requirements that improve control, usability or evaluation.

FR	Module	Requirement	Priority
FR-01	Auth	System shall authenticate users with credentials and issue a signed JWT carrying the user's role.	High
FR-02	Auth	System shall enforce role-based access (Student, Staff, Administrator) on every protected endpoint.	High
FR-03	Student	Authorised users shall create, view, update and delete student records holding student ID, roll number, name, department, year and section.	High
FR-04	Enrollment	System shall capture several face angles per student and store a deep embedding for every accepted sample.	High
FR-05	Enrollment	System shall reject enrollment samples that are blurred, too small, poorly illuminated or that contain more than one face.	High
FR-06	Detection	System shall detect multiple faces in a single camera frame.	High
FR-07	Recognition	System shall align each face, compute its embedding and compare it with the enrolled gallery by cosine similarity against a configurable threshold.	High

FR	Module	Requirement	Priority
FR-08	Recognition	System shall label faces below the threshold as UNKNOWN / INVALID STUDENT and shall not record them as attendance.	High
FR-09	Liveness	System shall evaluate liveness evidence and shall not mark attendance for a static presentation.	High
FR-10	Temporal	System shall confirm attendance only after repeated consistent observations of the same identity.	High
FR-11	Timetable	Administrator shall create and manage timetable slots with day, time, department, semester, section, subject, faculty and classroom.	High
FR-12	Session	System shall create an attendance session automatically when the current time falls inside a slot and shall prevent a duplicate session for the same slot and date.	High
FR-13	Session	Administrator shall be able to activate a session manually for testing and demonstration.	Medium
FR-14	Attendance	System shall store first-seen and last-seen timestamps for every student in every session.	High
FR-15	Student Portal	Student shall view profile, attendance history and attendance percentage.	Medium
FR-16	Staff	Staff shall monitor live attendance of active sessions and review session history.	High
FR-17	Reports	Staff and administrator shall export filtered attendance data as CSV, Excel and PDF.	High
FR-18	Audit	System shall log actor, action, target and reason for every administrative change to attendance or configuration.	High
FR-19	Settings	Administrator shall configure recognition and liveness thresholds.	Medium

FR	Module	Requirement	Priority
FR-20	Evaluation	Administrator shall run the Academic Evaluation module to compare the baseline and the proposed pipeline.	Medium
FR-21	Resilience	System shall return structured errors for missing sessions, invalid identifiers, empty galleries and unsupported requests rather than silently accepting invalid data.	Medium

Table 3.1: Functional Requirements

3.4 Non-Functional Requirements

The non-functional requirements in Table 3.2 describe the qualities the system must exhibit regardless of the specific function being performed. Each is paired with the approach by which the design satisfies it.

NFR	Category	Requirement	Approach
NFR-01	Security	All non-public endpoints shall require a valid token.	JWT verification dependency on every protected route
NFR-02	Security	Passwords shall never be stored in plain text.	Salted one-way password hashing
NFR-03	Privacy	Raw biometric exposure shall be limited; access to biometric data shall be restricted to authorised roles.	Embeddings and sample metadata stored server-side; role checks on enrollment and student routes
NFR-04	Accuracy	Wrongly accepting an unknown face shall be treated as more serious than deferring a genuine one.	Cosine threshold, quality gate, liveness and temporal confirmation
NFR-05	Performance	Live feedback shall remain responsive on commodity hardware.	Lightweight YuNet detector and ONNX inference on the CPU; latency and FPS measured by the evaluator

NFR	Category	Requirement	Approach
NFR-06	Reliability	A failed frame shall not terminate a session, and the scheduler shall be idempotent.	Per-frame error handling; existing-session check before creating a session
NFR-07	Configurability	Decision thresholds shall be changeable without code changes.	Values stored in system configuration and editable by the administrator
NFR-08	Maintainability	Biometric algorithms shall be testable independently of the web layer.	Separate vision modules, API schemas and database models
NFR-09	Usability	Each role shall see only the functions relevant to it, and recognition states shall be visible.	Role-specific navigation; PRESENT, VERIFYING and UNKNOWN states on the live screen
NFR-10	Auditability	Every administrative change shall be traceable.	Audit log with actor, action, target and reason
NFR-11	Data integrity	Deleting a student shall not leave orphaned biometric or attendance data.	Cleanup of related embeddings and attendance records in the delete workflow
NFR-12	Portability	The system shall run on a single ordinary machine without a GPU.	Local SQLite database and pretrained ONNX models

Table 3.2: Non-Functional Requirements

3.5 System Modules

The system is decomposed into the eleven modules summarised in Table 3.3. The decomposition follows the flow of an attendance session: users authenticate, administrators prepare the timetable and enroll students, the vision pipeline turns camera frames into verified identities, and the attendance, reporting and audit modules turn those identities into records that people can inspect.

Module	User Role	Key Functions	Technology
M1 — Authentication and RBAC	All	Login, token issue, route protection, role checks	FastAPI dependencies, JWT
M2 — Student Management	Staff, Admin	Create, view, update and delete student records	FastAPI, SQLAlchemy
M3 — Biometric Enrollment	Staff, Admin	Multi-angle capture, quality checks, embedding storage	OpenCV, ONNX models
M4 — Vision Pipeline	System	Detection, alignment, embedding and matching	YuNet, SFace/ArcFace, NumPy
M5 — Liveness and Temporal Verification	System	Blink and micro-motion analysis, multi-frame confirmation	Eye aspect ratio, temporal tracker
M6 — Timetable Scheduler	Admin, System	Slot management and automatic session creation	Python scheduler
M7 — Live Attendance	Student, Staff, Admin	Camera workflow, live states, attendance ledger	JavaScript, FastAPI
M8 — Dashboards and Portal	All	Role-specific views, history and percentage	HTML, CSS, JavaScript
M9 — Reporting	Staff, Admin	Filtered export to CSV, Excel and PDF	OpenPyXL, ReportLab
M10 — Academic Evaluation	Admin	Baseline versus proposed metrics	Python, NumPy, OpenCV
M11 — Audit and Configuration	Admin	Audit log and threshold settings	SQLAlchemy

Table 3.3: System Module Summary

3.6 Use Case Description

Student use cases include login, viewing the profile, starting attendance, reviewing attendance history and viewing the attendance percentage. Staff use cases include login, viewing active sessions, monitoring live attendance, managing students, reviewing history and exporting reports. Administrator use cases extend these operations to timetable configuration, subject and faculty management, system settings, academic evaluation and audit review. The principal use cases are described in Table 3.4.

ID	Use case	Actor	Description and outcome
UC-01	Log in	Student, Staff, Admin	User supplies credentials; system verifies them, issues a token and opens the dashboard for the user's role.
UC-02	View profile and attendance	Student	Student opens the portal to see academic details, attendance history and attendance percentage derived from stored records.
UC-03	Start attendance	Student, Staff	User opens the live camera workflow for the active session; recognised faces are shown with their state.
UC-04	Enroll biometrics	Staff, Admin	Staff captures several face angles; accepted samples are converted into embeddings and linked to the student.
UC-05	Monitor live attendance	Staff, Admin	Staff views PRESENT, VERIFYING and UNKNOWN states and first-seen and last-seen times for the active session.
UC-06	Review history	Staff, Admin, Student (own)	User filters past sessions and records by date, department or student.
UC-07	Export report	Staff, Admin	User selects filters and downloads the data as CSV, Excel or PDF.
UC-08	Manage timetable	Admin	Administrator creates and edits slots, subjects and faculty; the scheduler uses them to open sessions.

ID	Use case	Actor	Description and outcome
UC-09	Correct a record	Admin	Administrator changes an attendance record by supplying a reason; actor, target and reason are logged.
UC-10	Run academic evaluation	Admin	System compares the baseline and proposed pipelines on the selected dataset and reports the metrics.
UC-11	Review audit and settings	Admin	Administrator inspects the audit log and adjusts recognition and liveness thresholds.

Table 3.4: Use Case Descriptions

3.7 Data Flow Diagram

The primary data flow begins with a camera frame. The frame is passed through face detection and preprocessing. Detected faces are assessed for quality and aligned before deep embedding extraction. The resulting representation is compared with enrolled biometric records. A verified identity is passed to the temporal attendance layer, which updates the session ledger and timestamps. The frontend receives structured recognition and attendance status through the API. Figure 3.1 shows the Level 0 context diagram, in which the Smart Attendance System is the single process exchanging data with the student, staff and administrator actors, the camera, the attendance database and the report outputs.

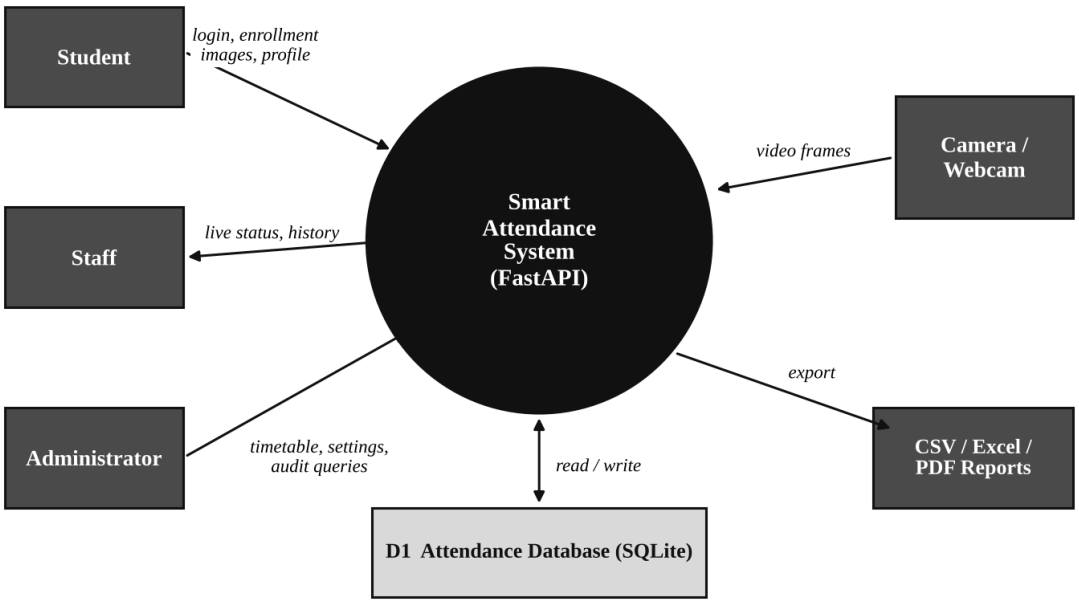


Figure 3.1: System Context / Data Flow Diagram — Level 0

Expanding the single process, the first sub-process authenticates a user and returns a token; the second manages student records and enrollment images; the third turns camera frames into identity candidates; the fourth maintains timetable slots and attendance sessions; and the fifth queries stored records to produce dashboards and reports. All five read and write the same attendance database, and none of them gives the browser direct access to it.

3.8 System Architecture

The system follows a three-tier architecture, shown in Figure 3.2. The presentation tier is a browser-based interface with separate student, staff and administrator views. The application tier is the FastAPI service, which hosts authentication and role checks, the timetable scheduler, the report generator, the academic evaluation module and the vision pipeline. The data tier consists of the SQLite database, accessed through SQLAlchemy, and the ONNX model assets used for inference.

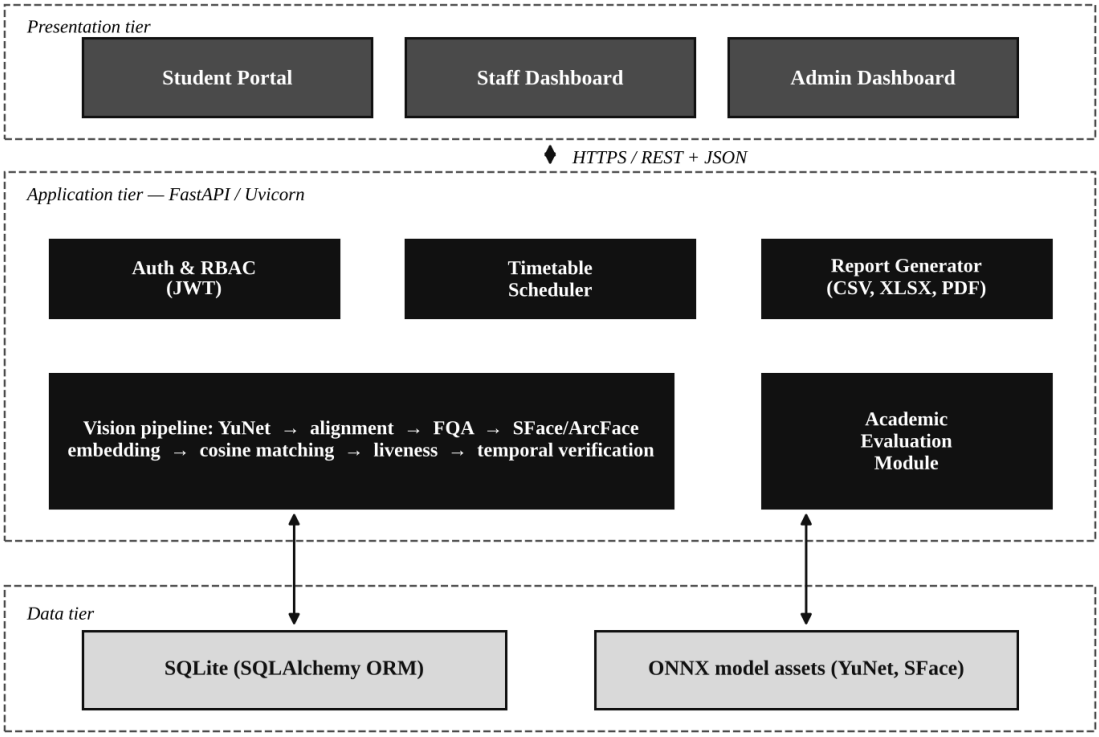


Figure 3.2: Smart Attendance System Architecture

Because the browser communicates only with the REST interface, the user interface can change without touching the recognition code, and the biometric modules can be tested without a browser. The scheduler runs inside the application tier and shares the same database session handling as the API routes, which keeps session creation consistent regardless of whether a session was opened by the clock or by an administrator.

3.9 Database Schema

The database contains entities for administrators, departments, subjects, faculty, students, face embeddings, timetable slots, attendance sessions, attendance records, audit logs and system configuration. Attendance records reference both a student and an attendance session. Biometric records store embedding-related information and sample metadata so that the enrolled gallery can be reconstructed. The principal tables and their roles are described in Table 3.5, and their relationships are illustrated in Figure 3.3.

Table	Important columns	Description
Student	student_id, roll_number, name, department, year, section, status	Central identity record for every enrolled student

Table	Important columns	Description
FaceEmbedding	id, student reference, vector, sample path, sample angle	One row per accepted enrollment sample; the stored gallery used for matching
Timetable	day, start and end time, department, semester, section, subject, faculty, classroom	A weekly class slot that the scheduler can turn into a session
AttendanceSession	date, timetable reference, start and end status	One concrete occurrence of a slot on a date
Attendance	student, session, status, first seen, last seen, liveness, source	The attendance ledger; one row per student per session
Subject / Faculty / Department	identifiers, names, codes	Reference data attached to timetable slots and students
AuditLog	actor, action, target type, target, reason, timestamp	Traceable record of administrative changes
SystemConfig	key, value	Recognition and liveness thresholds and similar settings

Table 3.5: Database Schema — Table Definitions

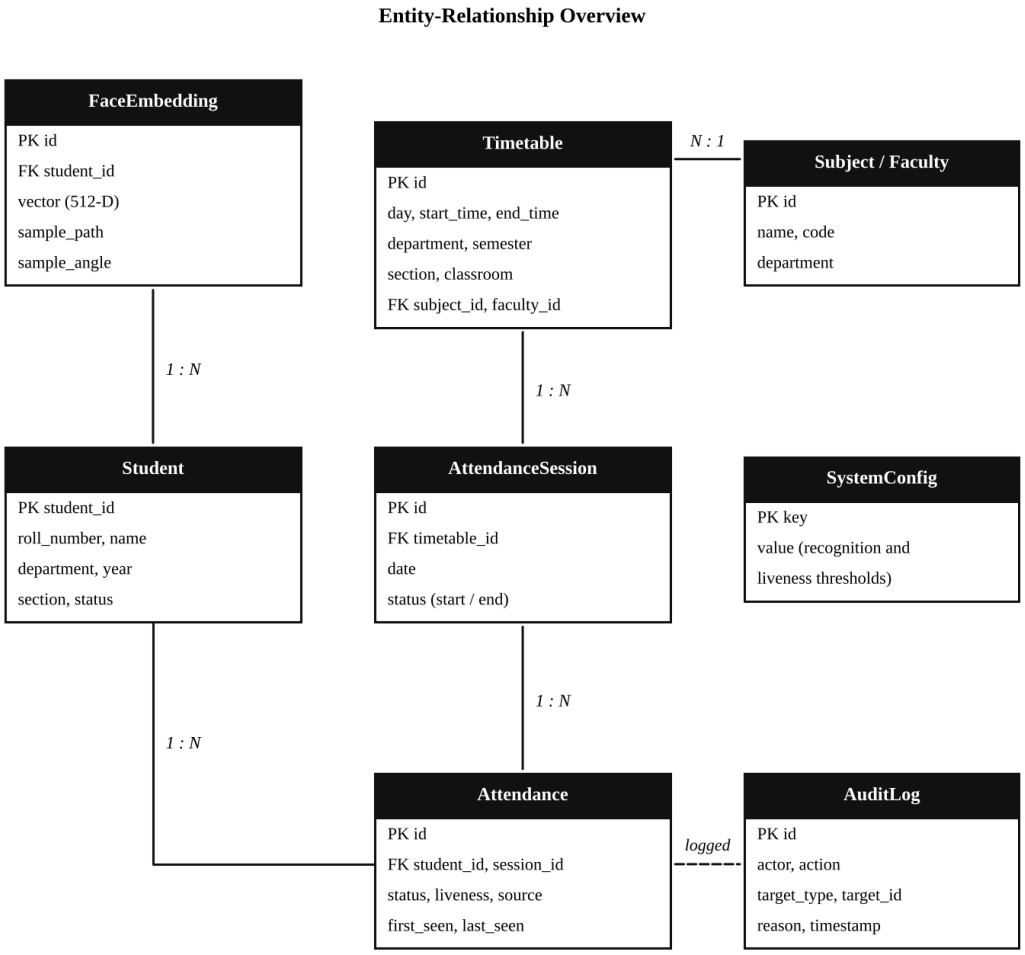


Figure 3.3: Entity-Relationship Diagram

3.10 Feasibility Study

Technical feasibility is supported by the use of Python, FastAPI, OpenCV, ONNX models and a local SQLite database, all of which are mature, documented and available without licence cost. The project does not require model training during each attendance session because the recognition pipeline consumes pretrained model assets and enrolled embeddings, and the only heavy computation is inference on a handful of cropped faces per frame.

Operational feasibility follows from the browser interface and the role-specific workflows. Students and staff need no installation and no special training, because the interaction consists of standing in front of a camera or opening a dashboard. Economic feasibility is improved by using commodity cameras and CPU-compatible software components; deep biometric inference performance depends on the deployment hardware, but

no specialised accelerator is required for the system to operate, and a larger deployment can later move to a faster machine without changing the software.

3.11 Risk Analysis

The principal risks to correct and trustworthy operation, their likelihood and impact, and the mitigation built into the design are summarised in Table 3.6.

Risk	Likelihood	Impact	Mitigation
Poor lighting or camera angle lowers recognition quality	High	Medium	Multi-angle enrollment, face-quality gate, configurable thresholds, camera placement guidance
Impersonation using a photograph or screen	Medium	High	Eye-aspect-ratio and micro-motion liveness; temporal confirmation; SPOOF_DETECTED state
Unknown person wrongly accepted	Low	High	Cosine threshold, quality gate, repeated observations before confirmation
Genuine student repeatedly rejected	Medium	Medium	Deferral instead of rejection on poor quality; threshold calibration; manual correction with audit
Unauthorised alteration of attendance records	Low	High	Role-based access, elevated permission for corrections, audit log with reasons
Duplicate or overlapping attendance sessions	Medium	Medium	Existing-session check for slot and date before creation
Incorrect or missing timetable data	Medium	High	Administrator-only timetable editing; manual activation for verification

Risk	Likelihood	Impact	Mitigation
Slow processing on weak hardware	Medium	Medium	Lightweight detector, CPU-compatible ONNX models, latency and FPS measurement
Loss or leakage of biometric data	Low	High	Server-side storage of embeddings, token-protected routes, encrypted storage listed as future work
Evaluation results over-interpreted	Medium	Medium	Published figures kept separate from measured ones; threats to validity stated in Chapter 7

Table 3.6: Risk Analysis

3.12 Assumptions and Dependencies

The system assumes that a camera is available and positioned so that student faces can be captured at a usable scale. It assumes that students are enrolled before live attendance begins and that timetable information is configured correctly. The recognition and liveness thresholds are configurable system values and should be calibrated for the deployment environment. Table 3.7 lists the assumptions and the consequence if each fails.

Assumption	Consequence if it does not hold
A camera with adequate resolution faces the area where students appear	Faces are small or blurred and are deferred by the quality gate; attendance is incomplete
Every student is enrolled with several clean face samples before the first session	Unenrolled students are shown as UNKNOWN / INVALID STUDENT and are not recorded
The timetable is complete and correct	Sessions open at wrong times or not at all; recognition results are not recorded
Thresholds are calibrated on representative local data	Either genuine students are rejected or look-alikes are accepted
Pretrained ONNX model assets are present on the server	The vision pipeline cannot start; a fallback detector is used only for detection

Assumption	Consequence if it does not hold
The server clock is accurate	Slot matching and first-seen / last-seen times become unreliable

Table 3.7: Assumptions and Their Consequences

The report also assumes that the project source represents the intended implementation. Where the conference paper describes a different algorithmic component, the implemented project is treated as the implementation truth.

CHAPTER 4
SYSTEM DESIGN

4.1 Backend Architecture

The backend is organised around FastAPI routes, Pydantic schemas, SQLAlchemy database models, authentication utilities, a timetable scheduler, report generation and a set of vision modules. This modular structure keeps API contracts separate from biometric algorithms and database persistence. The responsibilities of the main source modules are as follows.

- **main** — API routes and application orchestration, including the start-up of the scheduler.
- **auth** — password handling, token creation and verification, role dependencies and administrative action logging.
- **database** — SQLAlchemy models, session handling and lightweight migration helpers.
- **scheduler** — maps the current day and time to a timetable slot and creates or reuses the attendance session.
- **reports** — builds CSV, Excel and PDF documents from attendance queries.
- **vision package** — face detector, aligner, quality assessor, embedder, matcher, liveness checker, temporal tracker and academic evaluator.

4.2 Frontend Architecture

The frontend uses a single-page application shell with role-specific navigation sections. Student, staff and administrator views are shown according to the authenticated role. The interface includes dashboards, camera workflows, enrollment forms, timetable views, history tables, reports, academic evaluation and settings. Because it is written in plain HTML, CSS and JavaScript, it needs no build step and can be served directly by the backend or by any static file server.

The frontend stores the token returned at login and sends it with every request. It never reads the database or the model files directly; all biometric decisions arrive from the backend as structured JSON containing the detected faces, their states and the attendance status, and the interface merely renders them.

4.3 Security Design

Security is enforced at the API boundary. A user logs in with credentials; the backend verifies the password against a stored hash and issues a signed JSON Web Token [21] that carries the user's identity and role. Every protected route declares a dependency that verifies the token and, where required, checks the role. A missing or invalid token produces an authentication error, and a valid token with an insufficient role produces an authorisation error, as shown in Figure 4.1.

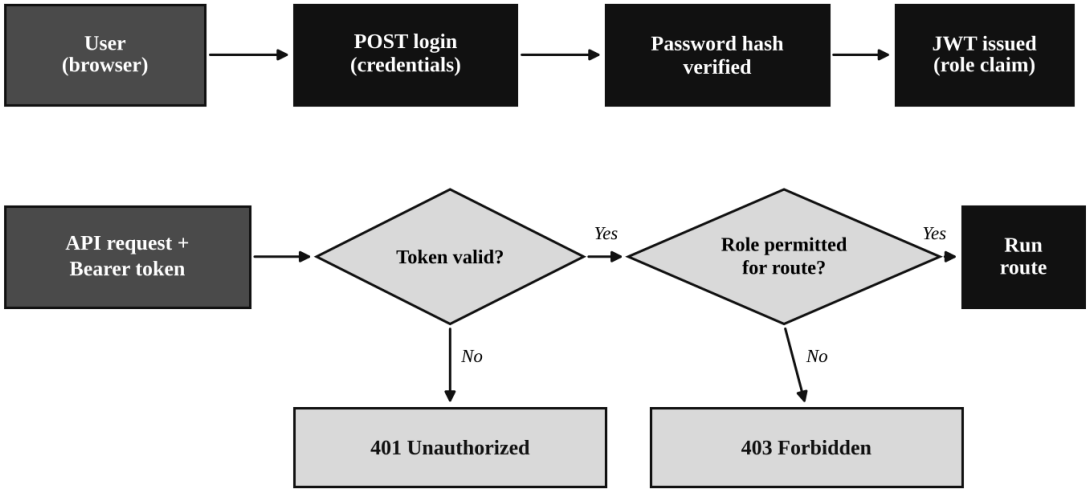


Figure 4.1: Role and Security Flow

Four design choices support this flow. Passwords are stored only as salted one-way hashes. Role checks are applied on the server for every sensitive route, so hiding a menu item in the interface is never the only protection. Administrative changes such as attendance corrections and configuration updates write an audit entry containing the actor, the action, the target and a reason. Finally, thresholds and secrets are held in configuration rather than in code. A production deployment would additionally place the service behind HTTPS and a reverse proxy, as discussed in Section 4.10.

4.4 REST API Design

The backend exposes a REST interface grouped by function. Table 4.1 lists the principal route groups, a representative operation for each, and the roles allowed to use them. Exact route names are defined in the backend source; the table describes the contract rather than reproducing every path.

Group	Representative operation	Method	Access
Authentication	Log in; read the current user	POST / GET	Public (log in); any authenticated user
Students	Create, list, view, update and delete student records	POST / GET / PUT / DELETE	Staff, Admin
Enrollment	Submit multi-angle face images for a student (enroll-faces)	POST	Staff, Admin
Timetable	Create and manage slots; list slots	GET / POST / PUT / DELETE	Admin (read: Staff)
Subjects and Faculty	Manage reference data	GET / POST / PUT / DELETE	Admin
Sessions	List active sessions; activate a session manually	GET / POST	Staff (list), Admin (activate)
Live attendance	Submit a camera frame for processing (attendance/process-frame)	POST	Authenticated user
Attendance history	List and filter attendance records	GET	Student (own), Staff, Admin
Attendance correction	Modify a record with a reason	PUT	Admin
Reports	Export attendance as CSV, Excel or PDF	GET	Staff, Admin
Academic evaluation	Run the evaluator and read its metrics	POST / GET	Admin
Settings and audit	Read and update thresholds; read audit entries	GET / PUT	Admin

Table 4.1: Representative REST API Endpoint Groups

Requests and responses are validated against Pydantic schemas, so malformed input is rejected before it reaches the database or the vision modules. Responses use consistent JSON structures and HTTP status codes, which allows the frontend to distinguish an authentication problem from a validation problem or a missing resource.

4.5 Database ER Diagram Description

As shown in Figure 3.3 (Chapter 3), the Student entity is the central biometric identity record. One student may have multiple FaceEmbedding rows because the implementation supports multiple sample angles. A Timetable slot can produce AttendanceSession records, one for each date on which the slot is active, and each session contains Attendance records for the participating students. A given student therefore has at most one attendance record per session, and that record carries the first-seen and last-seen timestamps and the liveness state.

AuditLog records connect administrative actions to actors and target entities, and SystemConfig holds the thresholds that influence recognition and liveness decisions. Reference entities such as department, subject and faculty are attached to timetable slots so that a report can show a session in the vocabulary the institution uses rather than as bare identifiers.

4.6 AI Service Design

The biometric component is a pipeline of independent stages, illustrated in Figure 4.2. Each stage has a single responsibility and a well-defined input and output, which makes it possible to replace the detector, the embedding model or the liveness rule without affecting the rest of the system. Table 4.2 summarises the stages.

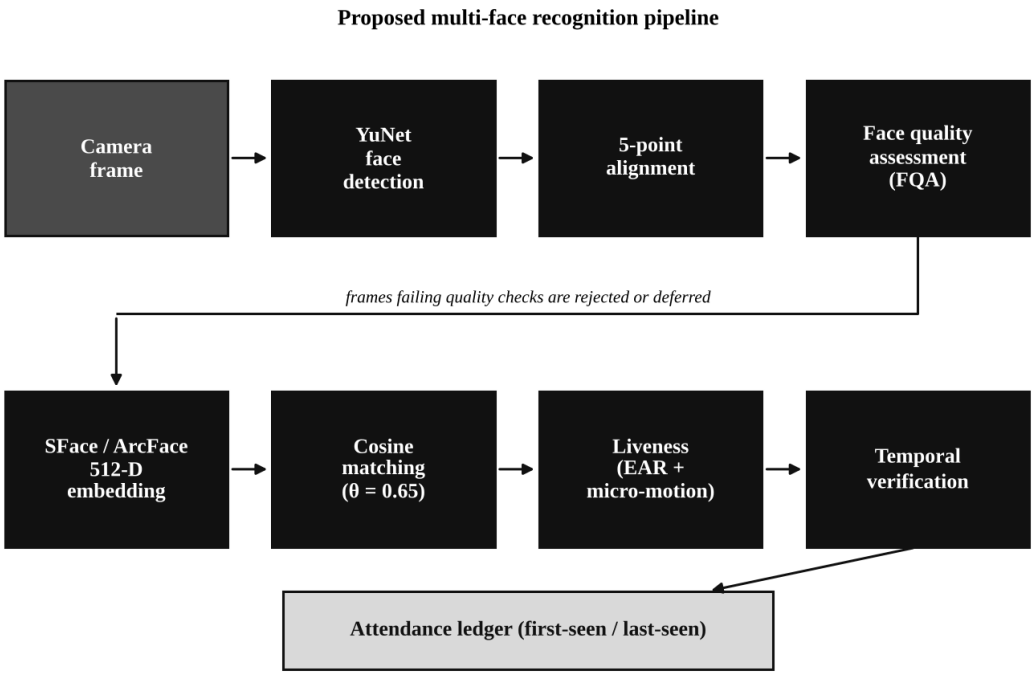


Figure 4.2: Proposed Face Recognition Pipeline

Stage	Input	Output	Purpose
Detection (YuNet)	Camera frame	Bounding boxes and five landmarks per face	Find every face in the frame, not only one
Alignment	Face and landmarks	Normalised face crop	Remove rotation and scale so embeddings are comparable
Quality assessment	Aligned crop	Accept, defer or reject	Block blurred, small, dark or multi-face samples
Embedding (SFace/ArcFace)	Accepted crop	512-D vector	Compact biometric representation
Matching	Probe vector, gallery	Best identity and cosine score	Compare against the threshold (0.65 by default)
Liveness	Landmark sequence	Live / not live evidence	Resist static photograph presentations

Stage	Input	Output	Purpose
Temporal verification	Repeated observations	Confirmed or pending identity	Prevent a single noisy frame becoming a record

Table 4.2: Vision Pipeline Stages

The pipeline uses pretrained models stored as ONNX assets. The recognition threshold of 0.65 for cosine-similarity decisions is a configuration value; matches above the boundary can proceed towards verification, while uncertain identities remain unrecognised. If the YuNet model cannot be loaded, a fallback detector keeps the detection stage available, although the report treats YuNet as the primary detector.

4.7 Sequence Flow

Figure 4.3 traces one live frame from the browser to the database and back. The browser posts the frame with its token; the backend validates the token and role, asks the scheduler for the active session, passes the frame to the vision pipeline, updates the first-seen and last-seen timestamps in the database and returns the faces, their states and the attendance status to the browser. The scheduler is the component that decides whether a session exists; the pipeline decides who is in the frame.

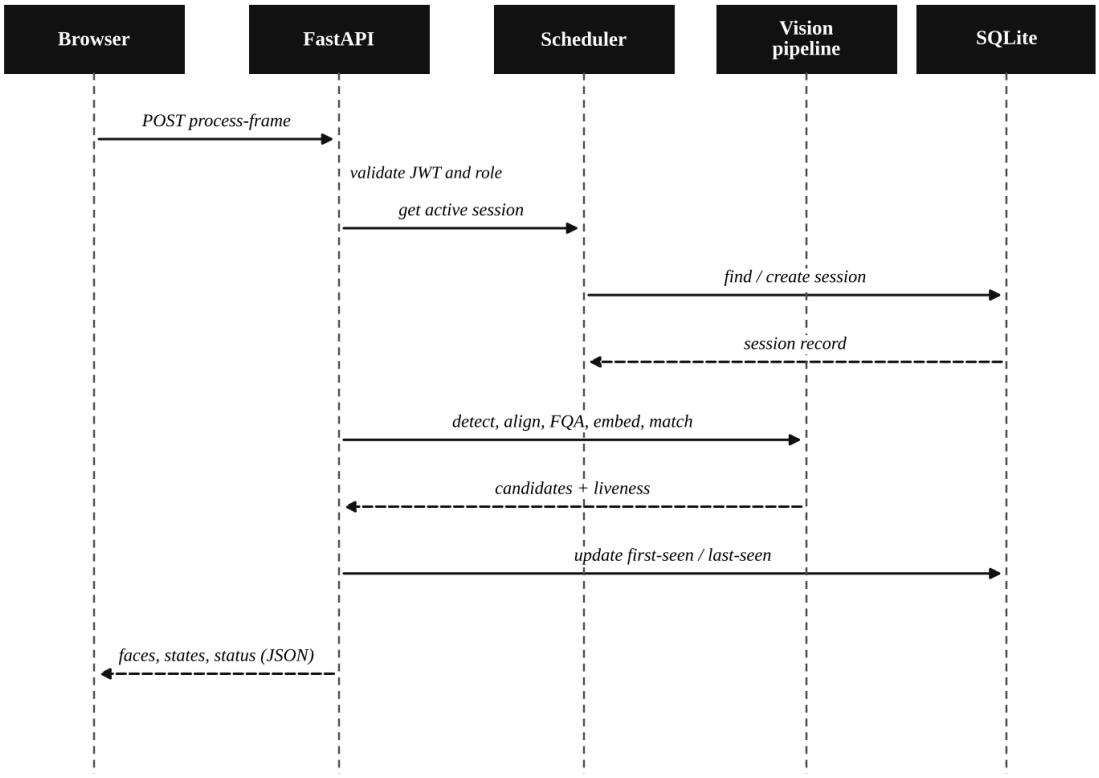


Figure 4.3: Sequence Flow of a Live Attendance Request

The session lifecycle that precedes this flow is shown in Figure 4.4. On every scheduler tick the current time is compared with the timetable. If the time lies inside a slot, the scheduler checks whether a session already exists for that slot and date and either reuses it or creates it, so a second session can never be opened for the same slot and date. Outside any slot, no session exists and recognition results are not written as attendance.

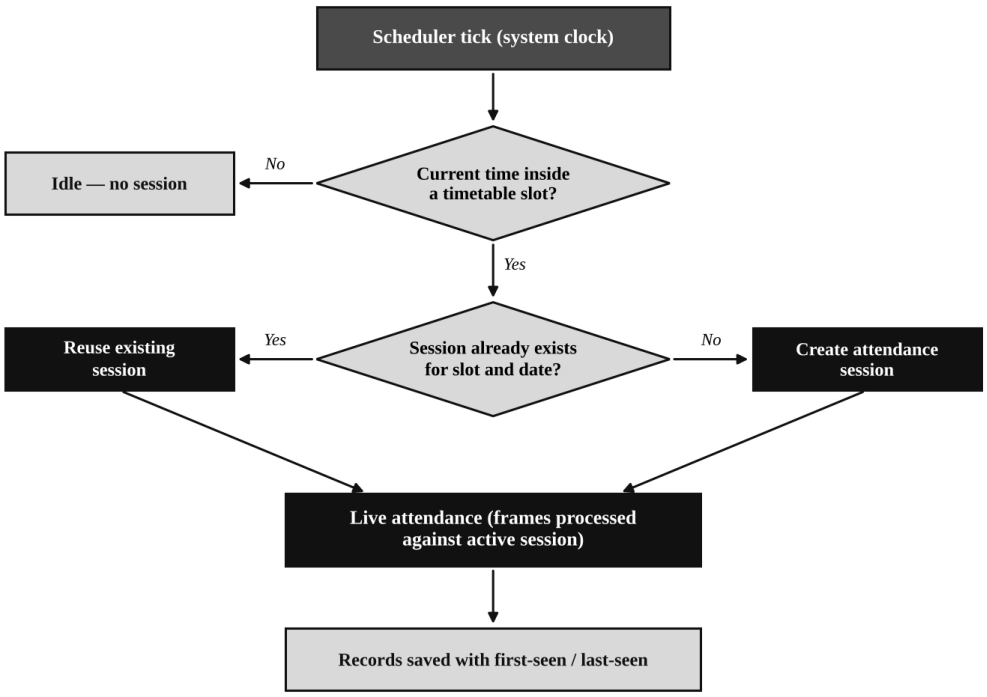


Figure 4.4: Timetable-Driven Attendance Session Flow

4.8 Error Handling and Resilience Design

Typical failure cases include missing active sessions, invalid authentication, duplicate student IDs, unavailable face models, insufficient face quality, unknown identities and invalid attendance edits. The API returns structured errors rather than silently accepting invalid data, and the scheduler checks for an existing session before creating a new one for a timetable slot. Table 4.3 lists the principal cases and the designed response.

Situation	Designed response
No active session for the current time	Structured error stating that no session is active; no attendance record is written
Missing or invalid token	Authentication error; no data returned
Valid token but insufficient role	Authorisation error; action not performed
Duplicate student identifier	Validation error; existing record unchanged
Face quality insufficient	Sample rejected at enrollment, or frame deferred during live attendance
Embedding model unavailable	Clear service error; no partial attendance written

Situation	Designed response
Empty enrolled gallery	All faces reported as UNKNOWN / INVALID STUDENT
Unsupported report request	Validation error with a meaningful message

Table 4.3: Error Cases and Designed Responses

4.9 Module Interaction Design

The frontend does not directly access the database or model files. It communicates through the FastAPI layer. The backend coordinates the scheduler, vision pipeline, database and report generator. This arrangement keeps interface changes independent of the recognition implementation and allows the biometric modules to be tested separately. The vision modules receive images and return plain data structures; they never import the web framework or open a database connection, which is what makes them straightforward to test and replace.

The scheduler and the live-attendance route interact only through the database: the scheduler creates sessions, and the route looks up the active session before it writes. This indirection avoids shared mutable state between the clock and the request path.

4.10 Deployment Architecture

For local deployment the project can be started with Uvicorn and accessed through a browser at the configured localhost address. The application stores data under the project data directory and uses local ONNX model assets, so a single machine with a camera is enough to demonstrate and evaluate the whole system. A production deployment would place the FastAPI service behind a secure reverse proxy with HTTPS, move secrets and persistent storage to managed infrastructure, replace SQLite with a server database if many concurrent users are expected, and encrypt biometric data at rest.

CHAPTER 5
IMPLEMENTATION

5.1 Authentication and RBAC Module

The backend provides login, current-user inspection, protected route dependencies and role checks. The application distinguishes administrator, staff and student workflows. When a user logs in, the submitted password is checked against the stored hash and a signed token containing the user's identity and role is returned. Every subsequent request carries that token, and each protected route verifies it through a shared dependency before any business logic runs. Figure 5.1 shows the login and role selection screen.

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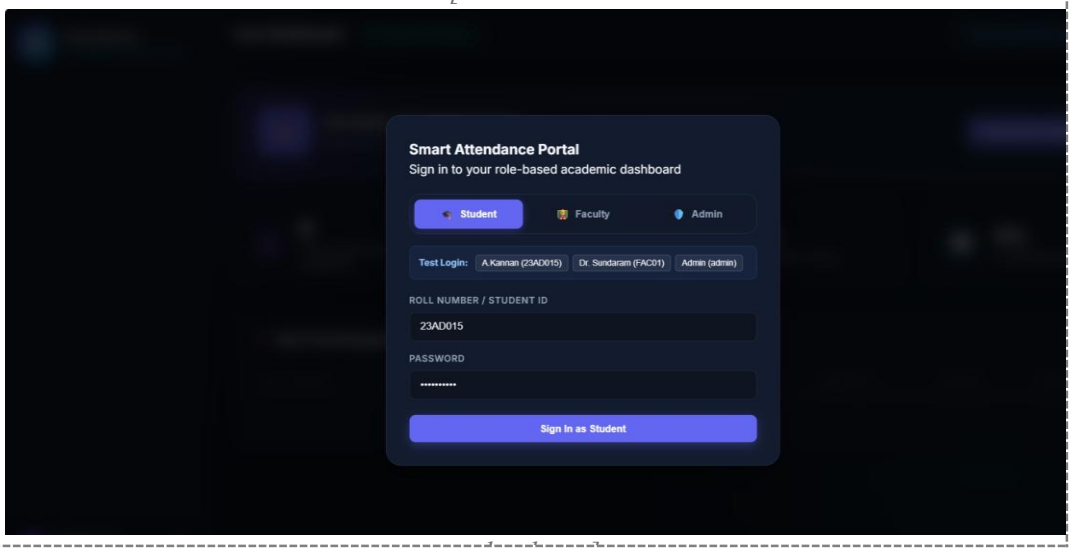


Figure 5.1: Login and Role Selection

The frontend navigation is dynamically organised around the role in the token, preventing ordinary users from seeing administration functions as their primary interface. The server-side checks nevertheless do not depend on the interface: a student who manually requested an administrator route would receive an authorisation error. Table 5.1 summarises the permissions granted to each role.

Function	Student	Staff	Admin
View own profile, history and percentage	Yes	Yes	Yes
Start live attendance for the active session	Yes	Yes	Yes
Monitor active sessions and live attendance	No	Yes	Yes

Function	Student	Staff	Admin
Manage student records	No	Yes	Yes
Enroll biometric samples	No	Yes	Yes
Review attendance history of all students	No	Yes	Yes
Export CSV, Excel and PDF reports	No	Yes	Yes
Manage timetable, subjects and faculty	No	View	Yes
Correct an attendance record	No	No	Yes
Run academic evaluation	No	No	Yes
Change thresholds; view audit log	No	No	Yes

Table 5.1: Role Permissions

5.2 Student Management

Student management supports creating, viewing, updating and deleting student records. The student record stores the identity and academic information required for attendance: student identifier, roll number, name, department, year, section and status. Duplicate identifiers are rejected by validation, so two records can never claim the same identity. Deleting a student also triggers cleanup of associated biometric and attendance data according to the backend workflow, so that no orphaned embeddings remain in the gallery after a student leaves. Figure 5.2 shows the student management and enrollment screen.

[Insert screenshot here]

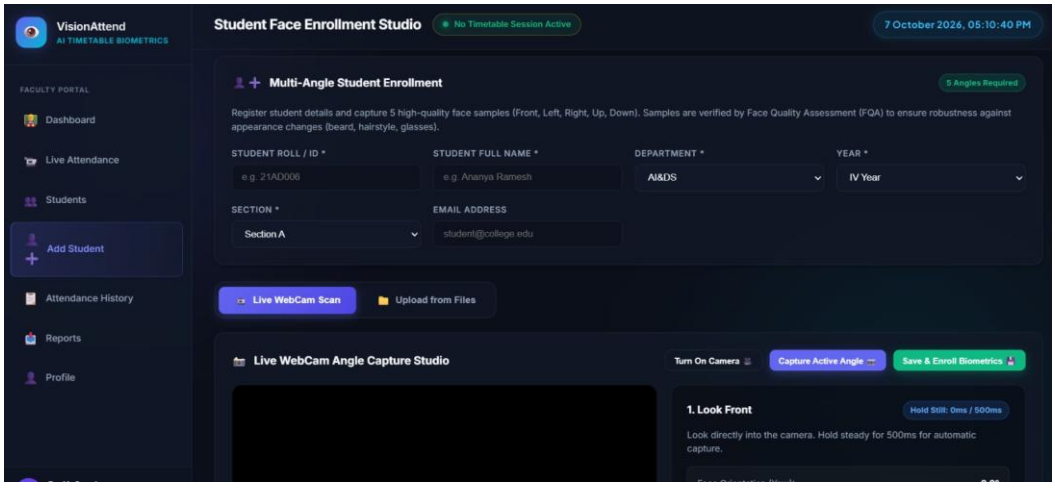


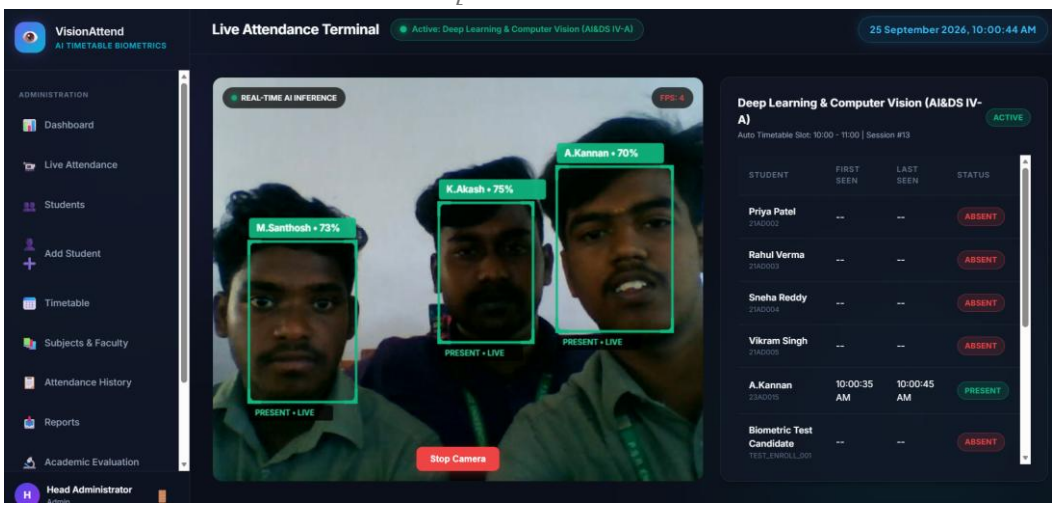
Figure 5.2: Student Management and Enrollment

5.3 Biometric Enrollment

Enrollment is an administrative procedure carried out before classes begin. After the student record exists, staff opens the enrollment screen and captures several face images in front of the camera, covering a frontal view and slight left, right, upward and downward poses. Capturing several angles is a deliberate design decision: a gallery built from a single frontal photograph tends to fail as soon as the student turns the head, and the literature reviewed in Chapter 2 identifies pose variation as a recurring weakness [3], [5].

Each captured image is processed by the same detection and alignment stages used during live attendance. The face-quality assessment then decides whether the sample may enter the gallery; a blurred, small, dark or multi-face crop is rejected and the operator is asked to capture it again. Accepted samples are converted into embeddings and stored with the student reference, the sample path and the sample angle, so that the enrolled gallery can be reconstructed at any time. Figure 5.3 shows the multi-angle enrollment screen.

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enshot here]

Figure 5.3: Multi-Angle Biometric Enrollment

5.4 Face Detection

The primary detector in the implementation is OpenCV YuNet [19], with a fallback detection path. The detector is designed for multiple faces rather than a single student at a time, which matters because students rarely enter a classroom one by one. Each detected face is represented by a bounding box and by associated landmark information, namely the eye centres, the nose tip and the mouth corners, which later stages use for alignment and liveness. Figure 5.4 shows how a frame is converted into aligned face crops.

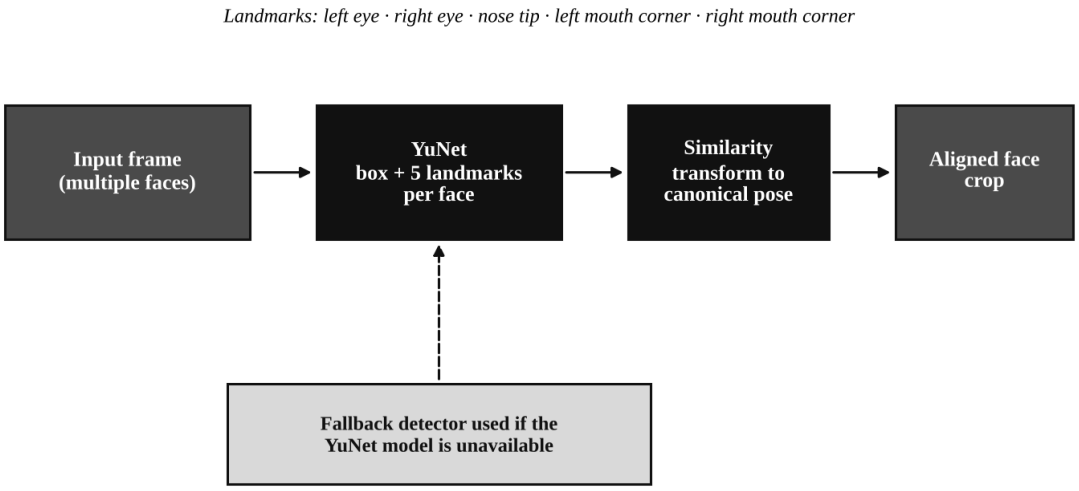


Figure 5.4: Face Detection and Alignment

YuNet was selected because it is small enough to run at interactive rates on an ordinary CPU, which keeps the system within the hardware assumptions of Chapter 3 and avoids the GPU cost that the multi-face tracking studies [6] identified as a barrier for institutions.

5.5 Face Alignment and Quality Assessment

Detected faces are normalised using a five-point similarity transformation toward the canonical ArcFace alignment geometry. Given the five detected landmarks, the transformation estimates a rotation, an isotropic scale and a translation that map them as closely as possible onto the template positions, and the face is warped accordingly. Alignment removes in-plane rotation and scale differences so that two images of the same person produce comparable embeddings.

Face-quality assessment checks properties such as blur, size, illumination and the presence of multiple faces in an enrollment crop. The purpose is to prevent poor samples from entering the biometric gallery and to reduce false triggers during attendance. The checks are intentionally conservative: during live attendance a frame that fails the quality gate is deferred, not rejected permanently, because the same student will usually be seen again a moment later in a better frame.

5.6 ArcFace Embedding

The implementation exposes an ArcFaceEmbedder abstraction and loads an ONNX deep face recognition model. The project documentation describes 512-dimensional deep embeddings. An ArcFace-style network [17] is trained with an additive angular margin so

that embeddings of the same person cluster tightly in angle while different people are pushed apart, and the SFace model [18] distributed with OpenCV follows the same principle. The embedding acts as a compact biometric representation that can be compared with stored enrollment vectors without retraining the network during attendance.

Because the network is used purely for inference, adding a new student requires only computing and storing new vectors. This is the same practical advantage that the conference paper [1] attributes to a non-parametric classifier: unlike a retrained softmax classifier, the gallery can grow without disturbing existing students.

5.7 Face Matching

The live pipeline performs similarity-based matching against the enrolled gallery. The similarity between a probe embedding u and an enrolled embedding v is the cosine of the angle between them,

$$\cos(u, v) = (u \cdot v) / (\|u\| \, \|v\|)$$

which takes the value 1 for identical directions and decreases as the faces become less alike. The system configuration includes a recognition threshold of 0.65 for cosine-similarity decisions. Matches above the configured confidence boundary can proceed toward verification, while uncertain identities remain unrecognised and are displayed as UNKNOWN / INVALID STUDENT. Because the threshold is stored as a configuration value, an administrator can make the system stricter or more permissive after observing its behaviour on local data, without changing the program.

5.8 Liveness Detection

The liveness module uses eye-aspect-ratio behaviour, micro-motion and combined evidence to distinguish a live presentation from a static presentation. The eye aspect ratio [20] is computed from six landmarks around each eye as

$$EAR = (\|p2 - p6\| + \|p3 - p5\|) / (2 \, \|p1 - p4\|)$$

and falls sharply when the eye closes, so a sequence of ratios reveals natural blinking. Micro-motion complements it by observing the small, continuous movements of a living face that a printed photograph does not show. A liveness status is retained with the attendance information so that a successful biometric match can be associated with the verification state

that produced it. A presentation judged static is reported as SPOOF_DETECTED and is not marked present.

5.9 Temporal Verification

Temporal verification prevents a single noisy frame from immediately becoming a final attendance event. The tracker evaluates repeated observations of a candidate identity and its liveness result across consecutive frames. This stabilises the recognition decision and reduces transient false detections caused by motion blur or a partially visible face. It also keeps the interface honest: while evidence is accumulating the face is shown as VERIFYING, and only when the evidence is consistent does the state change to PRESENT.

This design trades a short delay for reliability. The authors' baseline paper [1] reports a mean verification time of about six seconds and relies on the redundancy of a 30 frames-per-second video stream to compensate for individual detection misses; the temporal tracker uses the same redundancy, but as an explicit confirmation rule instead of an implicit property of the stream.

5.10 Timetable Scheduler

The scheduler continuously monitors the academic timetable and creates an attendance session when the current time falls inside a configured slot. It checks for an existing session for the same timetable slot and date, preventing duplicate sessions. The frontend also provides an administrative on-demand activation mechanism for testing, so that the whole workflow can be demonstrated outside real class hours.

The scheduler is the component that removes the need for any student action. In the QR-based systems discussed in Chapter 2, a session starts because a teacher generates a code and each student must respond; here a session starts because the clock reaches the slot, and the only action required of a student is to be present.

5.11 Live Attendance

The live attendance interface shows multiple recognised faces and their state. The visual states are confirmed PRESENT, VERIFYING and UNKNOWN / INVALID STUDENT, with a presentation judged static reported as SPOOF_DETECTED. The attendance ledger records first-seen and last-seen timestamps, allowing the system to retain evidence of when a student entered and was last observed. Figure 5.5 shows how a face moves between states and Table 5.2 defines each state.

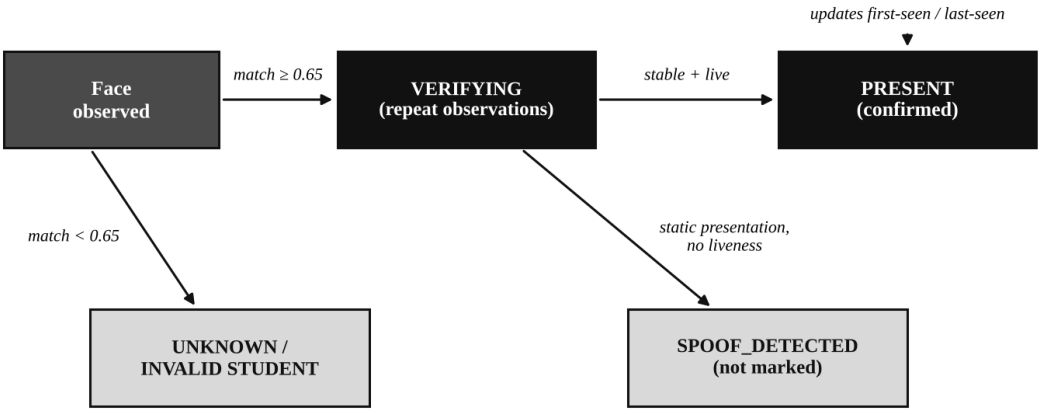


Figure 5.5: Live Attendance State Workflow

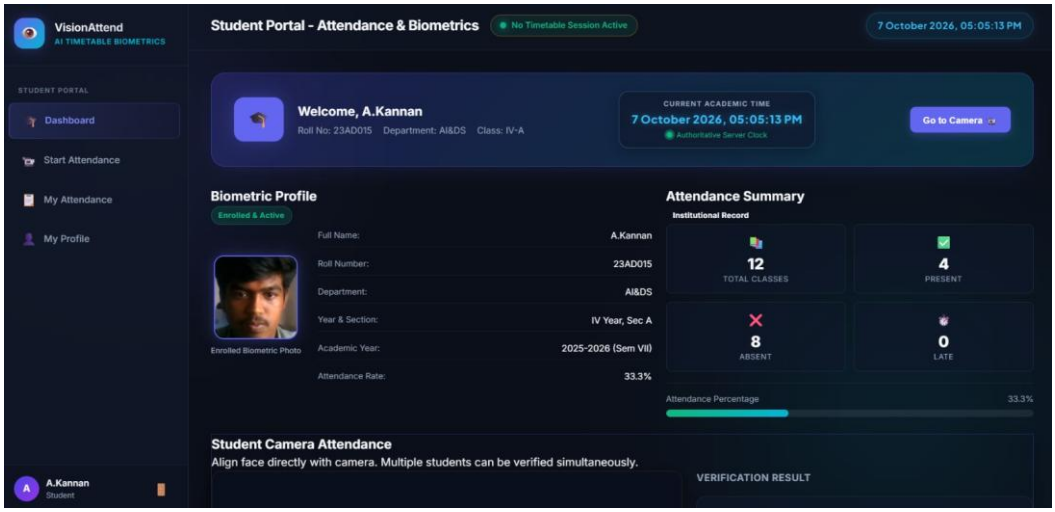
State	Meaning	Attendance recorded?
PRESENT	Identity matched above the threshold, live and stable across repeated observations	Yes — first-seen and last-seen updated
VERIFYING	Candidate identity seen but not yet confirmed by repeated observations	No — pending
UNKNOWN / INVALID STUDENT	Best similarity below the threshold, or no enrolled student matches	No
SPOOF_DETECTED	Match found but liveness evidence indicates a static presentation	No

Table 5.2: Live Attendance States

5.12 Student Portal

The student portal provides a dashboard, an attendance start workflow, personal attendance history and a profile view. The profile contains academic information and an attendance percentage derived from stored attendance records. Student-facing access is intentionally narrower than staff or administrator access, so a student can see only personal data. Figure 5.6 shows the student dashboard.

[Insert screens



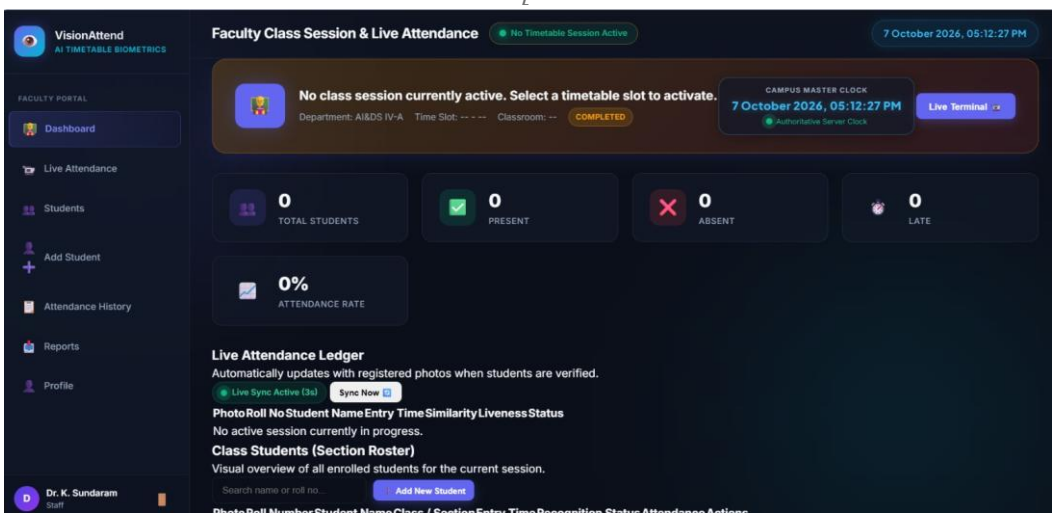
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Figure 5.6: Student Dashboard

5.13 Staff and Admin Dashboards

Staff views focus on active sessions, live attendance, student records, history and reports. Administrator views additionally include timetable management, subjects and faculty, academic evaluation, system settings and audit logs. This separation supports least-privilege access to institutional functions: a faculty member can see and export what is needed to run a class, but cannot alter the evidence on which the records rest. Figure 5.7 shows the staff and administrator dashboard.

[

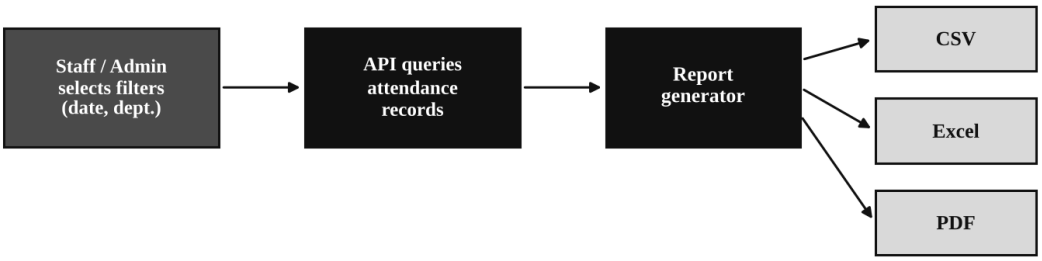


Insert screenshot here]

Figure 5.7: Staff and Administrator Dashboard

5.14 Reporting Module

The project includes report generation in three common formats. CSV provides raw tabular data, Excel provides structured spreadsheet output and PDF provides a print-oriented report. Filters such as date or department can be applied through the reporting workflow before export, as shown in Figure 5.8. Report generation only reads the stored records, so exporting never alters attendance data.



Exports are read-only: generating a report does not change stored attendance data.

Figure 5.8: Report Generation Workflow

5.15 Academic Evaluation Module

The academic evaluator compares a baseline pipeline with the proposed pipeline and computes accuracy, precision, recall, F1-score, false acceptance rate, false rejection rate, latency and FPS. The baseline is represented in code as Haar Cascade plus a 128-dimensional feature representation and Euclidean matching, while the proposed path uses YuNet face detection, quality processing, a 512-dimensional embedding and cosine similarity.

Because a real classroom dataset with many impostors is hard to collect, the evaluator constructs genuine probes from the enrolled images by applying lighting, blur and contrast variations, and it includes unknown or impostor cases. This controlled probe generation makes the comparison repeatable, and it also defines how the results must be interpreted, as discussed in Section 7.6. Figure 5.9 shows the evaluation workflow.

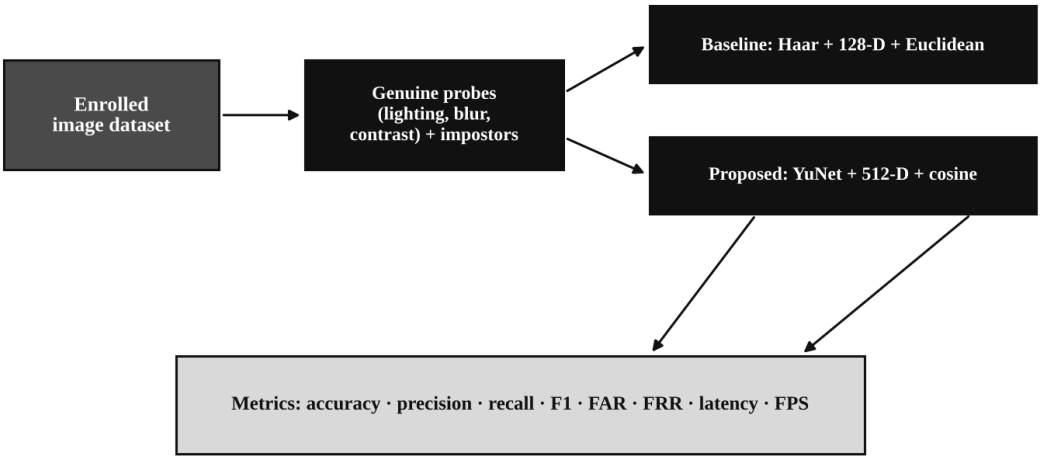
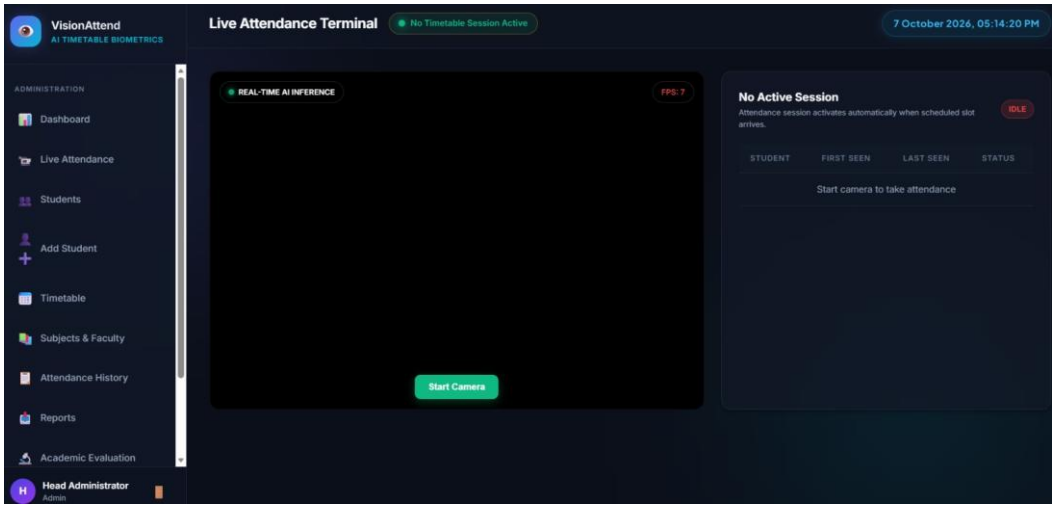


Figure 5.9: Academic Evaluation Workflow

5.16 Audit and Configuration

The database contains audit logs and system configuration. Recognition and liveness thresholds are stored as configurable values. Administrative changes can record the actor, target type, action and reason. This provides a traceable path for corrections and configuration changes rather than treating the attendance ledger as an unprotected spreadsheet. Figure 5.10 shows the audit and system settings screen.

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nshot here]

Figure 5.10: Audit Log and System Settings

CHAPTER 6
TESTING

6.1 Testing Strategy

The testing strategy combines unit-style checks for individual modules, integration checks for the API and database, role and security checks, browser-level workflow verification and biometric pipeline evaluation. Figure 6.1 shows the order in which the layers are verified, each layer being exercised before the next relies on it. The project contains multiple scratch and test scripts covering enrollment, role workflows, KNN matching, biometrics, session behaviour, audit operations and deletion flows.



Each layer is verified before the next layer relies on it.

Figure 6.1: Functional Testing Flow

The tables in this chapter state the expected behaviour for each test condition, and these expected results serve as the acceptance criteria for the system. Measured performance of the biometric pipeline is deliberately not asserted in this chapter; it is produced by the Academic Evaluation module and discussed, with its limitations, in Chapter 7. Table 6.1 summarises the testing layers.

Layer	Scope	What is verified
Module checks	Detector, aligner, quality assessor, embedder, matcher, liveness, tracker	Each stage returns valid output for normal, degraded and empty input

Layer	Scope	What is verified
API and database	Routes, schemas, models, scheduler	Valid requests persist correctly; invalid requests are rejected with structured errors
Role and security	Token and role dependencies, audit log	Each role reaches only its permitted functions; changes are logged
Workflow	Browser dashboards, enrollment, live attendance, reports	End-to-end scenarios behave as designed for each role
Biometric evaluation	Academic evaluator on enrolled data	Accuracy, precision, recall, F1, FAR, FRR, latency and FPS

Table 6.1: Test Suite Summary

6.2 Functional Test Cases

Table 6.2 lists representative functional test cases with the behaviour expected from the system. Each case can be run through the interface or through the scripted checks, and each corresponds to one or more functional requirements in Table 3.1.

ID	Module	Test case	Expected result
TC-01	Auth	Log in with valid student, staff and administrator credentials	Token issued; dashboard of the matching role opens
TC-02	Auth	Log in with an incorrect password	Authentication error; no token issued
TC-03	Student	Create a student with a new identifier	Record saved and visible in the student list
TC-04	Student	Create a student with an existing identifier	Validation error; existing record unchanged
TC-05	Enrollment	Enroll five clean face angles for a student	Five embeddings stored and linked to the student

ID	Module	Test case	Expected result
TC-06	Enrollment	Submit a blurred or very dark image	Sample rejected by the quality gate
TC-07	Enrollment	Submit an image containing two faces	Sample rejected; operator asked to retry
TC-08	Recognition	Present an enrolled student in good light	Face moves from VERIFYING to PRESENT
TC-09	Recognition	Present several enrolled students together	All faces detected and tracked independently
TC-10	Recognition	Present an unenrolled person	Shown as UNKNOWN / INVALID STUDENT; no record written
TC-11	Liveness	Hold a printed photograph of an enrolled student	Not marked present; SPOOF_DETECTED or no confirmation
TC-12	Temporal	Face visible for a single transient frame	No final attendance record
TC-13	Scheduler	Current time inside a slot	Session created for the slot and date
TC-14	Scheduler	Scheduler tick repeated in the same slot	No duplicate session created
TC-15	Attendance	Student seen early and again later in the session	First-seen and last-seen timestamps differ and are stored
TC-16	Reports	Export a filtered attendance report in each format	CSV, Excel and PDF files generated; data unchanged
TC-17	Audit	Administrator corrects an attendance record with a reason	Record updated; audit entry holds actor, action, target and reason

ID	Module	Test case	Expected result
TC-18	Delete	Delete a student with enrolled samples and attendance	Related embeddings and attendance cleaned up

Table 6.2: Functional Test Cases

6.3 Biometric Pipeline Testing

Biometric testing should cover normal frontal faces, slight left or right pose, slight up or down pose, blur, low illumination, partial occlusion, multiple faces and unknown persons. Enrollment tests focus on whether poor-quality images are rejected. Live tests focus on whether repeated observations stabilise before attendance is marked. Figure 6.2 shows which pipeline stage each test condition primarily exercises, and Table 6.3 states the expected behaviour.

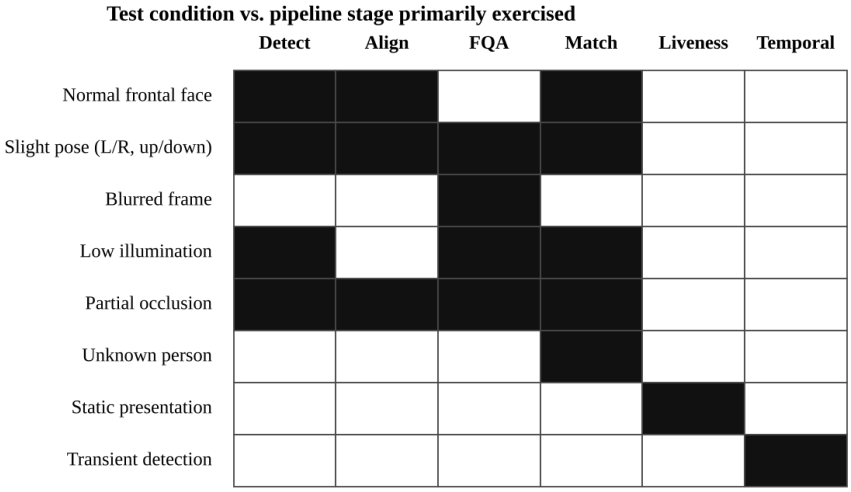


Figure 6.2: Biometric Test Conditions

Condition	Check	Expected behaviour
Normal face	Detection and embedding	Valid candidate
Slight pose	Alignment and embedding	Recognition remains possible
Blurred frame	Face quality assessment	Reject or defer
Low illumination	Quality assessment and matching	Defer, or match only if similarity stays above the threshold

Condition	Check	Expected behaviour
Partial occlusion	Detection and matching	Match only when similarity is above the threshold
Unknown person	Similarity threshold	UNKNOWN
Static presentation	Liveness	SPOOF_DETECTED / not marked
Transient detection	Temporal tracker	No immediate final mark

Table 6.3: Biometric Test Conditions and Expected Behaviour

6.4 Role and Authorization Testing

Role testing verifies that student, staff and administrator accounts receive only the operations appropriate to their roles. Sensitive endpoints such as student management, timetable modification, attendance correction and system configuration require elevated permissions. The checks include logging in as each role, attempting each restricted route with the wrong role and confirming that the response is an authorisation error, and confirming that the interface does not even show a restricted function to a role that may not use it.

6.5 Timetable and Session Testing

Timetable testing verifies correct slot selection by day and time, session creation, duplicate-session prevention and manual force activation. The scheduler should not create a second attendance session for the same configured slot on the same date. Boundary cases are examined deliberately: a time exactly at the start of a slot, a time just after its end, a day with no slots and two slots for different sections that overlap in time.

6.6 Reporting and Export Testing

Reporting tests verify that attendance records can be filtered and exported without changing the underlying data. CSV should contain raw fields, Excel should preserve tabular structure and PDF should present a readable institutional report. The same filter applied to the three formats must return the same set of records, and an export request with no matching records must produce an empty but valid document instead of an error.

6.7 Security and Isolation Testing

Security tests cover invalid credentials, expired or missing tokens, insufficient roles, unauthorised attendance modification and audit trail creation. Database-level relationships are

also checked to ensure that a student record, its biometric samples and attendance records remain internally consistent. Table 6.4 lists the principal security checks.

Test	Action	Expected result
Missing token	Call a protected route without a token	Authentication error
Invalid or expired token	Call a protected route with a tampered token	Authentication error
Student on staff route	Student requests student-management or report routes	Authorisation error
Staff on admin route	Staff requests settings, evaluation or correction routes	Authorisation error
Student reads another student's history	Request another student's records	Denied or limited to own records
Unauthorised modification	Attempt to change an attendance record as staff	Rejected; record unchanged
Audit trail	Administrator changes a record or a threshold	Audit entry created with actor, action, target and reason

Table 6.4: Security and Authorization Tests

6.8 Error Handling Testing

Error handling should be verified for missing timetable sessions, invalid student IDs, empty biometric galleries, poor-quality images, unknown faces and unsupported report requests. The system should fail with a meaningful state rather than silently marking attendance. In particular, a failure partway through processing a frame must not leave a half-written attendance record behind.

6.9 Requirements Traceability

Each major functional requirement can be traced to a backend route, service or frontend view. Authentication maps to the authentication module; biometric enrollment maps to the student enrollment endpoint; live attendance maps to frame processing; timetable

requirements map to the scheduler and timetable routes; reporting maps to the report generator. Table 6.5 gives the matrix.

Requirement	Implementation area	Test cases	Status
Authentication and roles (FR-01, FR-02)	backend/auth.py and auth routes	TC-01, TC-02	Implemented
Student CRUD (FR-03)	backend/main.py student routes	TC-03, TC-04, TC-18	Implemented
Biometric enrollment (FR-04, FR-05)	student enroll-faces route and vision modules	TC-05 – TC-07	Implemented
Live recognition (FR-06 – FR-10)	attendance/process-frame and vision modules	TC-08 – TC-12	Implemented
Timetable sessions (FR-11 – FR-13)	scheduler.py and session routes	TC-13, TC-14	Implemented
Timestamps (FR-14)	attendance ledger	TC-15	Implemented
Reports (FR-17)	reports.py	TC-16	Implemented
Audit and settings (FR-18, FR-19)	auth.py logging and configuration	TC-17	Implemented
Academic evaluation (FR-20)	vision/academic_evaluator.py	Evaluator run	Implemented

Table 6.5: Requirements Traceability Matrix

6.10 Manual Verification Walkthrough

A practical walkthrough used for the final demonstration proceeds in the following order.

1. Log in as an administrator and confirm that the timetable, subjects and faculty are configured.
2. Add a student and enroll the student with five face angles, confirming that poor samples are rejected.

3. Open the live attendance screen for the active session.
4. Present an enrolled face and confirm the recognised state and timestamps.
5. Present an unknown face and verify that it is not accepted.
6. Open the attendance history and export CSV, Excel and PDF reports.
7. Review the administrative action in the audit log and the system configuration.

CHAPTER 7
RESULTS AND DISCUSSION

7.1 System Performance Results

The project includes an empirical Academic Evaluation module that measures accuracy, precision, recall, F1-score, false acceptance rate, false rejection rate, latency and FPS. It constructs genuine probes using lighting, blur and contrast variations and includes unknown and impostor cases. The evaluator is therefore suitable for obtaining project-specific performance once the local enrolled dataset is selected. Table 7.1 defines the metrics, and the standard definitions in terms of true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) are given below.

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \qquad Precision = TP / (TP + FP)$$

$$Recall = TP / (TP + FN)$$

$$F1 = 2 \cdot Precision \cdot Recall / (Precision + Recall) \qquad FAR = FP / (FP + TN) \qquad FRR = FN / (FN + TP)$$

Metric	Purpose
Accuracy	Overall correct classification rate
Precision	Reliability of positive identity decisions
Recall	Ability to recognise genuine identities
F1-score	Balance between precision and recall
FAR	Rate of false identity acceptance
FRR	Rate of genuine identity rejection
Latency	Average processing time per probe
FPS	Approximate processing throughput

Table 7.1: Evaluation Metrics and Their Purpose

The authors' conference paper [1] reports measurements for its Haar, Dlib and ResNet-34 pipeline with a distance-weighted KNN classifier, and these published values form the baseline against which the implemented pipeline is positioned. They are summarised in Table 7.2. The paper reports 99.29% video recognition accuracy with a cumulative precision of

0.9947, recall of 0.9841 and F1-score of 0.9892 over 4,059 correctly localised video frames, and an average verification time of about 6.00 seconds across lighting and pose conditions. These values belong to the published paper and are cited here as literature evidence only; they are not measured results of the implemented system.

Measure	Published value
Haar Cascade detection rate — static images (3,000 samples)	58.33% (failure to detect 41.67%)
Haar Cascade detection rate — video frames (5,173 samples)	78.46% (failure to detect 21.54%)
Recognition accuracy — video frames (4,059 frames)	99.29%
Precision / Recall / F1-score (cumulative)	0.9947 / 0.9841 / 0.9892
Classifier comparison on video — SVM / MLP / weighted KNN	99.12% / 99.25% / 99.29%
Mean verification time — bright, frontal / dim, yaw tilt	3.80 s / 13.77 s
Composite mean verification time	6.00 s
Batch of six simultaneous students — total cycle / accuracy	822 ms / 97.2%

Table 7.2: Published Baseline Performance Reported in the Conference Paper [1]

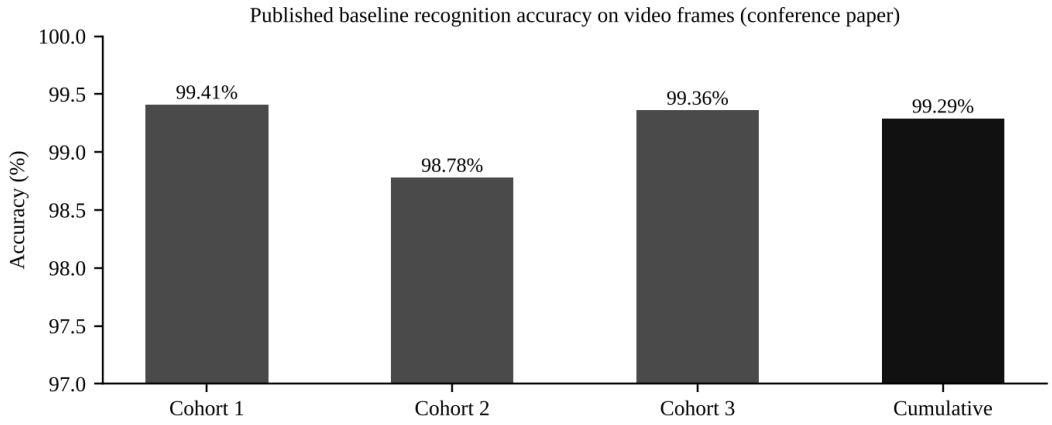


Figure 7.1: Published Baseline Recognition Accuracy by Cohort

The published results are informative in two ways. They show that a detection stage based on Haar features misses a large share of faces in static images and a smaller but still significant share in video, which is the weakness the implemented project addresses with a

modern detector and a quality gate. They also show that verification latency is strongly affected by lighting and pose, rising from under four seconds in bright frontal conditions to almost fourteen seconds in dim conditions with a yaw tilt. Because the implemented pipeline differs in its detector, embedding and matcher, the conference figures should be treated as an expectation to be tested, and the evaluator is the instrument for testing it.

7.2 Functional Outcomes

The project successfully integrates the core workflow needed for an institutional attendance product: role-based login, student management, multi-angle enrollment, multi-face live processing, timetable sessions, attendance timestamps, history views, reports and academic evaluation. The system is therefore broader than a standalone face-recognition demonstration. Table 7.3 summarises the outcomes delivered against the objectives of Section 3.2.

Objective	Outcome delivered
Contactless multi-face attendance	Several faces are detected and tracked in one frame without manual roll call or QR scanning
Multi-angle enrollment	Multiple accepted samples per student are stored as embeddings
Reject poor or suspicious frames	Quality gate and liveness check defer or reject unsuitable evidence
Embedding and similarity matching	512-D embeddings compared by cosine similarity at a configurable threshold
Timetable-driven sessions	Sessions open automatically from timetable slots without duplicates
First-seen and last-seen timestamps	Stored with every attendance record
Role-based workflows	Student, staff and administrator dashboards with server-side enforcement
CSV, Excel and PDF reports	Filtered export without modification of stored data

Objective	Outcome delivered
Measurable evaluation	Academic Evaluation module with eight metrics for baseline and proposed pipelines

Table 7.3: Functional Outcomes Against Objectives

The combination of biometric evidence and academic scheduling is particularly important. A recognised face is not sufficient by itself; it must be interpreted inside the correct timetable session and stored as an attributable attendance record, following the decision flow shown in Figure 7.2.

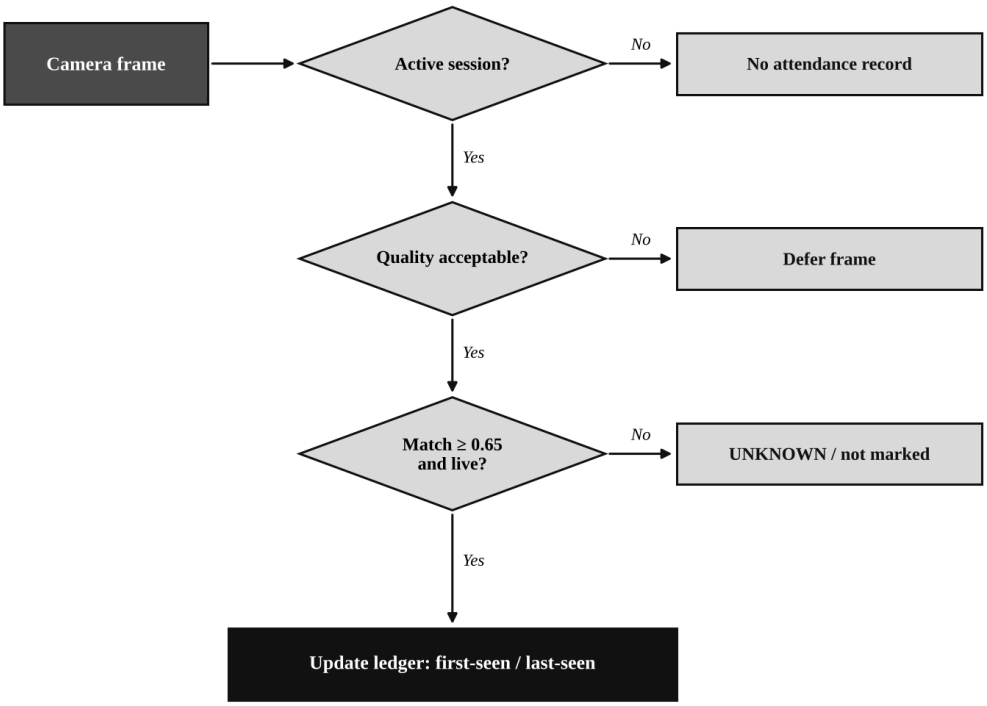


Figure 7.2: Attendance Outcome Flow

7.3 Existing vs Proposed System

Before an automated system, attendance in most classrooms is taken by manual roll call or paper sign-in, or by token-based schemes such as QR codes and fingerprint devices. Table 7.4 compares these approaches with the proposed system on the dimensions that matter to an institution. The statements about existing approaches follow the problems documented in the conference paper [1].

Dimension	Manual roll call / paper sheet	QR-code system	Fingerprint device	Proposed system
Physical contact	None	None (phone needed)	Required	None
Student action needed	Respond or sign	Find app, log in, scan	Queue and touch scanner	None beyond being present
Proxy / buddy risk	High	High — code can be shared	Low	Low — face and liveness evidence
Class time used	Substantial	Moderate	Moderate to high for large groups	Minimal — runs in background
Multiple students at once	No	Yes	No	Yes — multi-face frames
Timetable awareness	Manual	Teacher starts session	None	Automatic from timetable
Entry and last-seen evidence	No	No	Limited	First-seen and last-seen timestamps
Audit of changes	Weak	Limited	Device log	Actor, action, target and reason logged
Reports	Manual transcription	Export	Export	CSV, Excel and PDF

Table 7.4: Existing Systems vs Proposed System — Comparative Analysis

The pipeline-level difference between the authors' research baseline and the implemented project is shown in Figure 7.3. The implemented pipeline retains the idea of a staged, decoupled recognition design but replaces each stage with a more capable component and adds liveness and temporal verification, which the baseline paper lists only as future work.

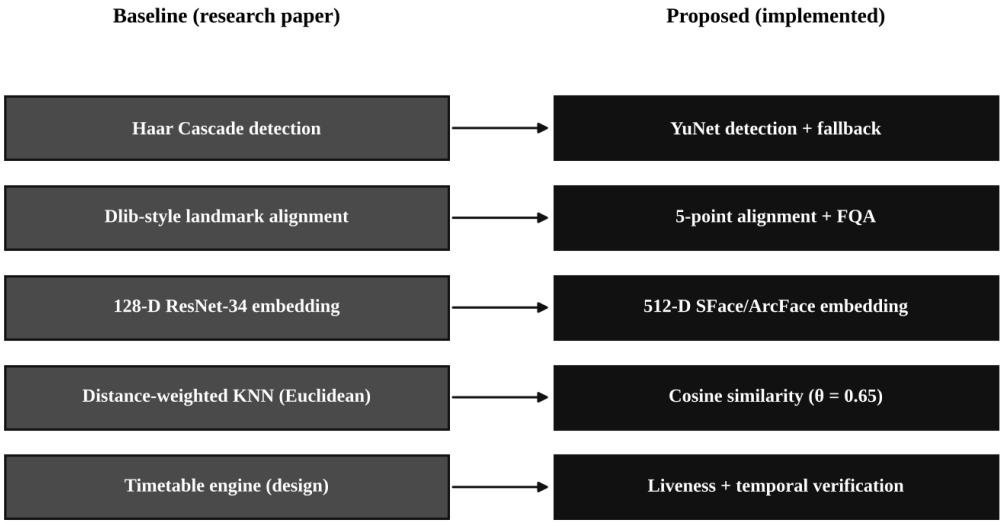


Figure 7.3: Baseline vs Proposed Pipeline Comparison

7.4 Discussion

The major design benefit is the integration of several moderate-strength controls rather than dependence on a single classifier. Multi-angle enrollment improves the gallery, face-quality assessment reduces bad samples, deep embeddings provide a stable representation, cosine similarity provides an explicit decision boundary, liveness adds presentation-attack evidence and temporal tracking reduces one-frame errors. No single stage has to be perfect, because the stages cover each other's weaknesses: a poor frame is stopped by the quality gate, a look-alike is stopped by the threshold, a photograph is stopped by liveness and a fleeting detection is stopped by the temporal rule.

The timetable scheduler further changes the meaning of recognition. The same face observed outside an active attendance session does not automatically become an attendance record. This reduces accidental marking and provides a clear operational boundary that manual and token-based systems do not have. It also removes the weakness of QR-based systems identified in the literature, since there is no code that can be forwarded to a friend.

The project also improves administrative usability. Student records, timetable slots, attendance sessions and reports are accessible through role-specific screens rather than requiring technical database access. Audit information gives administrators a traceable history for changes, so a disputed record can be explained by who changed it, when and why.

At the same time, computer vision performance remains environment-dependent. Camera placement, lighting, crowd density, pose, motion blur and enrollment quality can influence recognition, as the published baseline figures illustrate. For deployment, the thresholds should be calibrated using representative local data rather than copied blindly from another environment, and large lecture halls may need more than one camera, a point also raised in the conference paper [1].

7.5 Positioning Against Generic Alternatives

Compared with a QR-only system, the project removes the requirement for each student to find a code and present a phone. Compared with RFID or shared cards, the identity evidence comes from the person's face and not from an object that can be lent. Compared with a simple single-image face matcher, the proposed implementation adds quality, liveness and temporal layers. Table 7.5 compares the options across the capabilities that matter most.

Capability	**QR-only system**	**RFID / shared card**	**Single-image face matcher**	**Smart Attendance System**
Contactless	Yes	Tap required	Yes	Yes
Identity evidence tied to the person	No	No	Yes	Yes
Quality gate before decision	Not applicable	Not applicable	Usually no	Yes
Presentation-attack evidence	No	No	Usually no	Yes — EAR and micro-motion
Multi-frame confirmation	No	No	No	Yes
Timetable-driven sessions	Teacher-started	No	Usually no	Automatic
Role-based access and audit log	Varies	Varies	Usually no	Yes

Capability	QR-only system	RFID / shared card	Single-image face matcher	Smart Attendance System
Reports in several formats	Varies	Varies	Usually no	CSV, Excel, PDF

Table 7.5: Smart Attendance System vs Generic Alternatives

The system is best positioned as an institutional automation platform rather than as a claim of universal biometric accuracy. Its value comes from connecting recognition with academic workflow, authorisation and reporting.

7.6 Threats to Validity

Internal validity: performance measurements can be affected by dataset size, repeated images, synthetic impostor generation, camera hardware and threshold calibration. The project's academic evaluator itself documents a controlled probe-generation approach, in which genuine probes are produced from enrolled images by changing lighting, blur and contrast; therefore, its results should be interpreted as an engineering benchmark rather than a universal claim.

External validity: the conference paper's reported 99.29% result comes from its own experimental environment, with its own cohorts, camera and lighting, and should not be merged numerically with this project's future measurements. A fair final evaluation should use an independent test set with genuine and impostor identities representative of the target institution.

Construct validity: accuracy on probe images measures how well the pipeline separates identities, but an attendance system is judged by whether the right student is marked once in the right session. Quality, liveness, temporal confirmation and timetable control therefore have to be evaluated both separately, for example through ablation experiments, and together in a real classroom trial.

CHAPTER 8
CONCLUSION AND FUTURE WORK

8.1 Conclusion

The Smart Attendance System demonstrates a complete contactless attendance workflow built around multi-face recognition and academic timetable management. The implementation combines student enrollment, deep biometric representation, quality checks, similarity matching, liveness, temporal verification, session control, role-based access and institutional reporting.

The project moves beyond a simple recognition script by treating attendance as an auditable academic process. A student is enrolled before recognition, a session is created from the timetable, multiple observations are evaluated before attendance is finalised, and the resulting record can be reviewed and exported by authorised users. Each element addresses a weakness identified in the literature: Haar-based detection and Euclidean matching are replaced by YuNet and cosine similarity on deep embeddings, QR-based session starts are replaced by a timetable scheduler, and presentation attacks are met with liveness and temporal evidence.

The report has also kept published research measurements separate from the measurements the implementation can produce. The authors' conference paper provides the baseline and the research foundation, while the Academic Evaluation module provides the means of obtaining reproducible, project-specific figures on the institution's own data.

8.2 Future Enhancement

Future work can include larger institution-scale datasets, GPU-assisted inference, stronger anti-spoofing models, encrypted biometric storage, distributed database support, mobile-responsive staff interfaces, configurable institution-wide attendance policies and richer analytics. The most valuable extensions are summarised below.

- **Stronger anti-spoofing:** add infrared or depth sensing and learned blink or texture classifiers to resist replayed video and masks, as also proposed in the conference paper [1].
- **Scale and speed:** run several camera nodes with accelerated inference so that large auditoriums can be covered with low latency.

- **Data protection:** encrypt biometric embeddings at rest, define retention periods and move from SQLite to a managed server database.
- **Analytics:** use attendance history to identify students with chronic absence early and to report trends by subject, section and period.
- **Policy configuration:** allow institution-wide rules, for example a presence rule based on minimum time in class, to be configured per department.

Further research can evaluate recognition under real classroom crowding, extreme illumination and camera-angle variation. A carefully separated validation dataset can also quantify the effect of each component — YuNet detection, face-quality assessment, liveness and temporal verification — through ablation experiments.

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APPENDIX

A. Project Information

Project title: Development of a Multi-Facet Facial Recognition Model for Institutional Attendance Activities (implemented as the Smart Attendance System). Team: A. Kannan (95192309015), R. Muthu Bala (95192309028), M. Santhosh (95192309044) and K. Akash (95192309701). Guide: Mrs. V. Devinandhini, M.E., Assistant Professor, Department of Artificial Intelligence and Data Science, P.S.R. Engineering College, Sivakasi. Programme: B.Tech. Artificial Intelligence and Data Science, Anna University, Chennai.

B. Project Directory Structure

The representative project directory structure is given below. The listing names the principal source modules and their responsibilities.

Module	Responsibility
backend/main.py	API routes and application orchestration
backend/auth.py	Authentication, authorisation and administrative action logging
backend/database.py	SQLAlchemy models and database migration helpers
backend/scheduler.py	Timetable-driven session scheduler
backend/reports.py	CSV, Excel and PDF report generation
backend/vision/face_detector.py	YuNet and fallback face detection
backend/vision/face_aligner.py	Five-point facial alignment
backend/vision/face_quality.py	Face-quality assessment
backend/vision/face_embedder.py	Deep face embedding
backend/vision/matcher.py	Enrolled gallery matching
backend/vision/liveness.py	Liveness verification
backend/vision/temporal_tracker.py	Multi-frame stability
backend/vision/academic_evaluator.py	Baseline and proposed benchmark

C. Development Environment

Tool / Software	Purpose
Python	Backend, vision pipeline and evaluator runtime
FastAPI and Uvicorn	REST API framework and application server
OpenCV and NumPy	Image processing, YuNet detection and numerical operations
ONNX model assets (YuNet, SFace/ArcFace)	Pretrained detection and embedding inference
SQLAlchemy and SQLite	Object-relational mapping and local persistence
PyJWT-style token handling	Signed authentication tokens
OpenPyXL and ReportLab	Excel and PDF report generation
Web browser	HTML, CSS and JavaScript user interface

D. Configuration and Pipeline Parameters

Component	Setting	Notes
Face detector	OpenCV YuNet with fallback	Multi-face detection with landmarks
Alignment	Five-point similarity transform	Canonical ArcFace geometry
Embedding	512-dimensional ArcFace/SFace-style vector	ONNX inference; no training at runtime
Matching	Cosine similarity	Default recognition threshold 0.65, configurable
Liveness	Eye aspect ratio and micro-motion	Thresholds stored in system configuration
Enrollment	Five face angles per student	Quality-checked before storage
Authentication	JWT with role claim	Role checked on every protected route

E. Test Assets

Representative test assets in the project include scripts for enrollment verification, full-pipeline testing, staff audit and student workflows, role workflows, KNN matcher checks, deletion flows, biometric and session behaviour, and environment validation. The project data directory contains face-sample folders for enrolled identifiers. These samples demonstrate the intended multi-angle enrollment format and can be used to populate the biometric gallery before live attendance testing.

F. Recommended Final Demonstration Sequence

1. Login as administrator.
2. Verify department, subject, faculty and timetable configuration.
3. Add a student and capture five face angles.
4. Open the active attendance session.
5. Present multiple enrolled faces and verify recognition states.
6. Present an unknown face and verify rejection.
7. Review first-seen and last-seen timestamps.
8. Open attendance history.
9. Export CSV, Excel and PDF reports.
10. Run Academic Evaluation after the benchmark dataset is populated.
11. Review audit information and system configuration.

G. Glossary of Project-Specific Terms

Term	Meaning in this project
Enrollment	Capturing several quality-checked face samples of a student and storing their embeddings
Embedding	A 512-dimensional vector that represents a face and is compared by cosine similarity
Face-quality assessment	Checks on blur, size, illumination and face count that decide whether a sample or frame is usable

Term	Meaning in this project
Liveness	Evidence from eye-aspect-ratio behaviour and micro-motion that the face is a live presentation
Temporal verification	Rule requiring repeated consistent observations before attendance is finalised
Attendance session	One occurrence of a timetable slot on a date, to which attendance records belong
First-seen / last-seen	Timestamps of the first and the most recent confirmed observation of a student in a session
SPOOF_DETECTED	State shown when a match is found but liveness evidence indicates a static presentation

DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

Mrs. V. Devinandhini, M.E
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
devinandhini@psr.edu.in

Kannan.A
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
kannanandiappan695@gmail.com

Muthu Bala.R
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
balamuthu469@gmail.com

Santhosh.M
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
santhosh3122005@gmail.com

Akash.K
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
akashakash98047@gmail.com

ABSTRACT

Conventional techniques for tracking student attendance in educational institutions typically rely on the old-fashioned manual attendance method, the use of physical sign-in sheets, or simple biometrics such as fingerprint scanning. The methods employed in traditional attendance monitoring are known to be laborious, disruptive to lessons, vulnerable to human fraud, and unhygienic in the case of a large group of students. New developments in contactless technologies have made it possible to use QR codes with web-based applications but the novel method still has its issues including the need to avoid queues, the chance of fraud detection, and the requirement for someone to point a QR code this person's phone.

The Multi-Facet Institutional Attendance System (MFIAS) is being introduced in this paper as a contactless attendance system that is dependent on schedule which works all problems present in conventional systems. The manual scanning of people during attendance is not applicable any more thanks to the Automatic Departmental Timetable Engine which is scheduling classes and monitors them according to it. Final identity verification is performed by means of the multi-faceted computer vision that considers many real world possible hindrances like dynamic lighting in classrooms, various head position angles (tilt ± 15 degrees), and many more things that may block the identification (like eyeglasses and beard). The process employs the usage of Haar Cascade framework for the image detection process, Dlib 68 points model for landmarks issues, Residual Neural Network RegNet-34 for defining facial features, and KNN (k=3) method.

Additionally, it contains asynchronous multi-student batch checking, precise date and time recording for arrival and departure as well as duration calculating. The attendance management software features strict authorization at the administrator level which makes it possible to update attendance record only by the appropriate people. Extensive experiments show that the video classification from still pictures as well as from the live and previously recorded video gives the accuracy of 99.29 percent, 0 percent false omission rate and 6 seconds average verification time despite the changes in videos. Also, the cloud attendance software is able to produce tamper-proof log files and convert them into CSV, Excel and PDF format files that helps assure the safety of attendance taking process through biometrics.

I. INTRODUCTION

Attendance records in schools are considered to be vital part of education management. Such records are known for helping in various internal evaluations, approval to conduct research in laboratories, and accreditation of examinations in educational establishments. Nonetheless despite numerous innovations in modern-day technologies participation is rather old-fashioned and is taken mainly in shape of roll call when people are asked to fill out sheets with signatures or go through the process of using gadgets for collection of information about attendance such as fingerprint scanners. Despite the high

popularity of the methods that are used in this area they cannot be called efficient because roll calls may take up to 10-18 percent of class time; paper sheets are still subject to the dangers of buddy signing, and gadgets require physical contact and cannot be used in all situations.

To overcome these challenges, creative approaches were introduced in the form of mobile applications and QR codes scanning systems. In the traditional QR code approach, a teacher generates a code which has a period after which it expires and a student scans it to show that he is present. However, research shows that QR codes did not eliminate cheating, they just made it different since students can taking pictures of a QR code on the screen and send it to their friends outside the university. Also, the necessity for students to locate and open the application, log in to the network using the application and point the camera at the screen introduces additional difficulties and delays into the process of teaching.

Biometric facial recognition technology represents the primary way of performing academic identity verification without direct contact. After capturing human facial features, they cannot go missing or be altered; therefore, biometric facial recognition technology allows automatically keeping track of students in class just by watching them coming into the classroom. However, ready-to-use face recognition systems perform significantly worse in free unstructured educational environments. In ordinary classrooms students' faces may present a great variety of recognizable and unrecognizable features.

- **Changing Brightness:** Daylight differs from shining sunlight from the outside to dim light inside classrooms.
- **Tilting in Movement:** Students do not walk straight when passing through the doors of classrooms.
- **Facial Overlapping:** Common accessories of people, for example, eyeglasses with heavy glass, different types of beards and hats cover some important geometric reference points.
- **Delay in the Process of Ingress:** The usage of complex end-to-end implementation models causes significant costs resulting in the delay of data ingressing into the classroom.

This study develops an automated, contactless Multi-Facet Facial Recognition Attendance System to eliminate the challenges associated with existing limitations without requiring students to scan QR codes. The system comprises five important parts:

1. The Timetable-Based Punctuality System allows direct access to the department's timetable which allows to automatically plan attendance sessions without relying on QR codes.

2. Advanced Biometric Algorithms allow creating sophisticated data processing systems involving Haar Cascade classifiers, Dlib's 68-point face modeling method, ResNet-34, and K-Nearest Neighbor algorithms.

3. The usage of Simultaneous Student Verification allows the acquisition of information from the video stream and recording of the concurrent presence of learners.

4. Dynamic Audit of Entry and Exit Times enables registration of the time of arrival and departure of students from the classroom, creating the complete record of their presence in class.

5. Admin Security prevents anyone else from performing writing and reporting operations within the database.

The rest of the paper is organized into sections. The following section describes the previously conducted research and the methods which form the basis for the technique described in this paper. This section is anything but a general outline of the mathematical approaches and methods utilized in the realization of the project. The section IV gives the evaluation of the technique and includes the practical assessment of the technology used. The next section V contains the evaluation of the technical part of the system as well as organizational issues. Finally, section VI gives the conclusions and recommendations for the future research.

II. RELATED WORK & RESEARCH FOUNDATIONS

Over the last 20 years, automated attendance systems have come a long way. In the early days, these systems made use of the Viola-Jones object detection algorithm together with either Local Binary Pattern Histograms (LBPH) or Principal Component Analysis (PCA) to overcome the challenges related to capturing localized facial features. While these systems do not require heavy computational resources and allow the implementation of attendance systems on microcontrollers, they fail in extreme lighting situations, slight head movements and obscured faces.

Automated attendance systems have made great progress in the last 20 years. At first, this technology was based on Viola-Jones object detection algorithm combined with Local Binary Pattern Histograms (LBPH) and Principal Component Analysis (PCA). Hence, the algorithms did not need much power and, therefore, were usable even on microcontrollers, but they still did not manage to produce reliable results in difficult lighting conditions, in particular, when head movements or obscured faces were present.

The emergence of deep convolutional neural networks (CNNs) enabled researchers to switch to high-dimensional representation methods of analysis. Pattnaik and Mohanty conducted an evaluation of cloud-based computer vision services such as Google Cloud Vision or Amazon Rekognition that proved to have high accuracy in picture analysis. When switching to real-time video transmission the disadvantages of such methods became obvious: video-steaming creates heavy traffic load and also has untrusted latency. Moreover, in the case, where privacy is an issue, it requires stable connection to the Internet of the university's camera. According to the study performed by Lateef and Kamil, it became possible to track many people's faces in one smart classroom with the help of different YOLOv7 models.

Nevertheless, the fact that depth detection devices must operate at real-time frame rates necessitates the need for expensive GPU technology, making these products impractical for use in schools, as well. So as to cope with a certain kind of blocking of the face, Kamil and his co-authors came up with a system for online authentication that is capable of detecting individuals that have protective masks as required by the condition of the pandemic, achieving an efficiency of 81.8% in identification of such individuals. Nevertheless, this system is not good for solving the case of complicated distortions and is aimed only at solving the problem of identifying the lower part of the face with the nose covered with a mask. In addition, Afiyanto and his co-authors substantiated the effect of variation of illumination on the metric Euclidean distance which is reduced by 34% within the classrooms that are constantly illuminated by 150 lux and less.

The research by Jack and others into the field of multi-dimensional analysis of attendance management has resulted in a major breakthrough that solves many existing problems. According to the claims of Jack et al., the decoupled hybrid solution composed of Haar Cascade detection, Dlib landmark alignment, ResNet-34 128-dimensional vector encoding and distance-weighted KNN classifier shows 99.29% recognition rate in video streams and 6.00 seconds average verification time, while the other deep learning models are encountered with the obstacles caused by the shortcomings of the inference process.

TABLE I. Comparative Summary of Automated Attendance Literature

Reference	Core Methodology	Primary Advantage	Identified Limitations & Gaps
Bairagi et al. (2021)	Haar + LBPH	Minimal cost; fast writes.	Fails under off-pose & lighting shifts.
Trivedi et al. (2022)	OpenCV + Eigenfaces	Automates SQL logging.	Rigid alignment; zero occlusion tolerance.
Kamil et al. (2023)	SVM + Mask CNN	Masked face verification.	Isolated occlusions only; no pose handling.
Lateef & Kamil (2023)	YOLOv7 Deep Model	Continuous multi-face tracking.	Heavy GPU load; high latency at edge nodes.
Afiyanto et al. (2025)	Dynamic Filter + CNN	Illumination normalization.	High overhead; fails on composite angles.
Jack et al. (2026)	Haar + ResNet-128D + KNN	Multi-facet robustness (99.29%).	6.00s latency; lacks timetable automation.
Proposed MFIAS	**Timetable + ResNet + KNN**	**Zero QR; batching; In/Out logs.**	**Needs calibrated entry camera placement.**

III. PROPOSED SYSTEM ARCHITECTURE & METHODOLOGY

The Multi-Facet Institutional Attendance System or MFIAS has been developed with six modules that are interrelated to each other in a unique way. The modules are; (i) Student Biometric Enrollment Layer; (ii) Departmental Timetable Scheduling Engine; (iii) Multi-Students Computer Vision Ingestion Layer; (iv) Multi-Facet Deep Feature Embedding and Classification Engine; (v) In/Out Temporal Tracking and State Machine; and (vi) Role-Based Admin Governance and Multi-Format Export Module.

A.Student Enrollment & Biometric Blueprint Synthesis

Enrollment can be viewed as an administrative procedure that takes place before the operation of classes begins. Every student accepted by the institution is registered with his/her demographic details such as Roll Number, Name, Department, Year, and Academic Section. Enrollment is carried out in front of the camera, which captures 30 different images of the student's face.Frontal gaze under ambient and supplemented lighting.

- Yaw motion involves tilting forward and backward (by 15°).

• The study would consider the impact of wearing and not wearing glasses along with different types of facial hair.

The images indexed are processed through bilinear interpolation to achieve a regulated affine alignment of 400 × 400 pixels in size. All images are then converted to single-channel 8-bit grayscale format, thus effectively capturing the texture and geometry while disregarding color variations. Then, facial landmark points are extracted, converted to 128-dimensional embedding vectors by the ResNet-34 deep learning backbone, and stored in an indexed binary file through Pickle available in the Python environment. This way, the need for deep learning training in real time has been eliminated due to the pre-computed embeddings.

B. Automated Departmental Timetable Engine

A significant advancement brought about by MFIAS is the removal of the reliance on QR codes and mobile scans inside the elements. The attendance of students should adhere to the established formal schedule of classes. The Timetable Engine functions as an independent daemon thread interacting with the university database. The engine relies on the master schema of relationships:

```
TIMETABLE_SLOT = {
"slot_id": UUID,
"department": "AI&DS",
"semester": 6,
"section": "A",
"day_of_week": "MONDAY",
"start_time": "09:00:00",
"end_time": "09:50:00",
"grace_period_mins": 10,
"course_code": "AD8601",
"course_title": "Computer Vision & Deep Learning",
"assigned_faculty_id": "FAC_9519_01"
}
```

The hardware clock is compared to the timetable index all the time:

1. Pre-Session Trigger: In advance of the time of the session by 2 minutes, the system starts the camera pipeline, sets buffer streams, and saves the biometric vectors of students who are enrolled in the class from the respective department and semester.

2. Effective Presence Window: During the period from initial time to final time, video feeds are analyzed to assess student presence. Students verified during grace minutes are marked as Present; verifications occurring after the grace time are designated Late.

3. Completion of Session: After end time, the camera feeds are automatically disabled. Any student participating in the group who was absent during that period is logged as Absent. The final attendance log is saved.

C. Multiperson Face Recognition and Alignment

For the whole time of the session, the camera output is recorded at a rate of 30 frames per second (fps). Unlike the single portal cases, it should be noted that in reality the students do not walk alone. The platform uses OpenCV pre-trained frontal Haar Cascade classifier for the processing of the video stream. The system uses integral image representation to assess rectangular Haar-like features through a multi-scale sliding window:

$$ii(x, y) = \sum_{x' \le x,\, y' \le y} i(x', y') \quad (1)$$

In this notation, ii(x,y) refers to the cumulative value associated with facial recognition, while i(x',y') indicates the pixel intensity. The methodology is based on a series of AdaBoost classifiers eliminating undesirable image backgrounds in the earlier stages of analysis. All the bounding boxes Bk = [xk, yk, wk, hk] are acquired at once.

To achieve the required consistency despite head inclinations up to ±15°, the bounding box coordinates are aligned to the facial features of the person. A Dlib ensemble of regressor decision trees is used to determine the location of 68 key facial features including chin, eyebrows, nose, eyes, and mouth. The transformation between the inner corners of the eyes (features 37 and 46) and the tip of the nose (feature 31) is calculated and is used to transform the face to a frontal position.

D.Deep Feature Extraction & ResNet-34 128-D Metric Encoding

Affinities of normalized facial regions are learned using the Residual Neural Network (ResNet-34) architecture. Instead of performing identity classification like a softmax cross-entropy classification problem, our idea is to use the network purely as a feature extractor and convert facial landmarks based on geometry into 128-dimensional Euclidean space:

$$f: I \to \Re^{128} \quad ext\{such\ that\} \quad \|f(I)\|_2 = 1 \quad (2)$$

The ResNet-34 architecture has shortcut identity connections over 34 convolutional layers, allowing us to train deep networks without the problem of gradient vanishing. The weights are trained with the aid of the Triplet Loss method. The triplet Loss function makes it necessary for similar facial representations (anchor f(xia) and positive case f(xip) to be grouped close to each other and meaning that different facial representations (negative case f(xin)) must be separated from each other with a predetermined distance α:

$$L = \sum_i^N max(0, \|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha) \quad (3)$$

In our case, the value of α is equal to 0.2. By using this approach, the variations of the similar person facial appearance caused by eyeglasses, beard, or bad lighting create a minimal shift in the vector that is still located within the same identity area but separate from the identities of

other students because of the high distance in Euclidean space.

E. Distance-Weighted K-Nearest Neighbors (KNN) & Unknown Rejection

For live identity assignment, the normalized 128-dimensional vector x undergoes classification against enrollment symbols recorded in the database using the distance-weighted K-Nearest Neighbors algorithm with k = 3. The Euclidean distance calculated between this test vector and the training vector xj can be written as:

$$d(x, x_j) = \sqrt{(\sum_{m=1}^{128} (x_m - x_{j,m})^2)}$$

In order to minimize the impact of outliers caused by changing lighting conditions, the voting weights wj are assigned inversely proportional to the square of the Euclidean distance to the enrolled individual:

$$w_j = 1 / (d(x, x_j)^2 + \varepsilon)$$

Where ε = 10-5 avoids division by zero. The predicted identity y is found through the majority voting mechanism:

$$y = argmax_c \sum_{j=1}^{k} I(y_j = c) \cdot w_j$$

The Unknown Person Protocol: In order to avoid the cases of false positive identification, strict limitations regarding the Euclidean distance threshold are established.

```
IF min(d(x, x_j)) <= 0.60:
    Identity = y (Valid Student
Verified)
ELSE:
    Identity = "UNKNOWN / INVALID
STUDENT"
    Trigger Alert Flag & Log
Unregistered Face Capture
```

Temporal In/Out State Machine & Session Tracking

Contrary to single-instance check-ins, the requirement of educational accountability entails checking the time spent by each student inside the classroom. The MFIAS database captures 2 separate time stamps for individual students for each session.

1.In-Time (Entry Timestamp): This time stamp is generated as soon as a student is detected in the ongoing session for the first time:

entry_timestamp = CURRENT_TIMESTAMP.

2. Out-Time (Exit Timestamp): The last time stamp is updated every time a student is detected in the last 10 minutes of the session or when he/she leaves the class: exit_timestamp = CURRENT_TIMESTAMP.

3. Duration Calculation: Once the session is over, the duration is calculated as follows: duration_minutes = (exit_timestamp - entry_timestamp)/60

4. Attendance Status:

○ If entry_timestamp ≤ (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as PRESENT.

○ If entry_timestamp > (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as LATE.

○ If it is neither detected nor duration_minutes < 0.50 × session length it is recorded as ABSENT.

G. Role-Based Administrative Governance & Multi-Format Reports

It is important for the security integrity to ensure that neither faculty members nor students can modify attendance records. MFIAS also uses JWT authentication-based RBAC.

● Student/Public Role – Read-only access to personal attendance percentages.

● Faculty role – Read-only access to session summaries and attendance records in the classroom.

● Admin role – Only one who has the ability to reset attendance flags (such as being on duty). Administrative changes are recorded through explanation codes whereby the ID of the administrator is also logged in the system.

For archiving purposes, the website has embedded report generators that automatically generate reports in three different formats.

● CSV – Spreadsheet format with raw data.

● XLSX – Tables with colors coded status stamps and statistics of attendance.

● PDF – Printed reports with signatures.

IV. Experimental Setup, Results & Performance Evaluation

A.Experimental Environment & Dataset Curation

Real-life testing in an academic environment was simulated in a laboratory setting. Technology used in the computational testbed included an Intel Core i7 CPU, 16GB DDR4 RAM, and NVIDIA GeForce RTX 3060 GPU (for the ResNet embedding) and the processing of the CPU was employed for answering the question of whether edge computing is feasible.

The experimental data were collected utilizing the method presented by Jack et al. (2026). The images were created with various experimental conditions in mind (illumination (bright vs dim), head position (facing forward, tilted to the right or left at a 15-degree angle), clothing (with

or without glasses and beard). The evaluation included more than 3000 still images, over 5173 video frames, and real-life tests of continuous student entry.

B. Detection Stage Evaluation: Failure to Detect (FTD) Dynamics

System evaluation requires decoupling face detection from classification. Table II outlines the detection efficacy of the pre-trained Haar Cascade classifier across static frames and continuous video.

TABLE II. Haar Cascade Face Detection Performance Metrics

Evaluation Modality	Total Samples	Detected Samples	Detection Rate (%)	FTD Rate (%)
Static Image Dataset	3,000	1,750	58.33%	41.67%
Continuous Video Frames	5,173	4,059	78.46%	21.54%

It was noted that static photographs had an FTD of 41.67%, partly as a result of acute head positions and dim areas of low contrast that did exceed gradient limits set by Haar-like filters. In contrast, during regular video transmission (30 fps), the FTD dropped to 21.54%. Importantly, an FTD event does not result in a false acceptance; it simply rejects a frame that is unsuitable for further processing. Due to the temporally redundant nature of live broadcasts, a passerby walking through a doorway will yield anywhere from 60 to 120 frames for accurate detection as the student adopts a natural position.

C. Evaluation of Multilevel Recognition Performance

To evaluate the quality of classification, 4,059 frames correctly localized over video recordings were examined. The particulars of precision, recall, and F1-score for each student group are presented in Table III.

TABLE III. Recognition Performance Metrics on Video Frames

Student Identifier	Frames	Precision	Recall	F1-Score	Accuracy (%)
Student Cohort 1	1,215	0.9959	0.9934	0.9946	99.41%
Student Cohort 2	492	0.9979	0.9593	0.9782	98.78%
Student Cohort 3	2,352	0.9903	0.9996	0.9949	99.36%
Cumulative Average	**4,059**	**0.9947**	**0.9841**	**0.9892**	**99.29%**

The system demonstrated that the 128-dimensional ResNet-34 embeddings yield good separation between different classes while remaining close within classes regardless of the type of accessory or slight motion blur, since the overall recognition accuracy achieved by the system was 99.29%.

A. Classifier Testing: Comparison of KNN and Other Machine Learning Models To confirm the appropriateness of using distance-weighted KNN as the architecture, the same 128-dimensional embedding vectors were used to assess the classification accuracy of a Linear Support Vector Machine (SVM) and a Shallow Multi-Layer Perceptron (MLP with a hidden layer of 64 neurons running in Adam optimization mode).

TABLE IV. Comparative Benchmarking of Lightweight Classifiers

Classifier Head	Static Acc (%)	Video Acc (%)	Precision	Recall	F1-Score	Retraining Need
Linear SVM	99.94%	99.12%	0.9937	0.9749	0.9842	Full Retrain
Shallow MLP (64)	99.66%	99.25%	0.9937	0.9812	0.9931	Iterative BP
Weighted KNN (Ours)	**99.94%**	**99.29%**	**0.9947**	**0.9841**	**0.9892**	**Instant Insert**

Despite their accuracy scores being only slightly below that of KNN (99.12% and 99.25% for SVM and MLP respectively), in practice, the algorithms of both of these models should be re-tuned across the entire student base whenever a new student appears. In this regard, the distance-weighted KNN is a non-parametric algorithm that allows for an unrestricted addition of new students into the search space without affecting the speed of inference. *Response Latency & Multi-Facet Operational Robustness The real-time working capacity of the system was analyzed in different conditions to determine its response speed and False Omission Rates (FOR). The data is provided in Table V.*

A.TABLE V. Average Response Time (Seconds) Under Multi-Facet Variance

Illumination Setting	Head Orientation	With Glasses / Beard	Without Glasses / Beard	Avg Latency (s)
Bright (>450 lux)	Frontal Straight (0°)	3.79 s	3.81 s	3.80 s
Bright (>450 lux)	Yaw Tilt (±15°)	4.37 s	4.11 s	4.24 s
Dim (<150 lux)	Frontal Straight (0°)	7.84 s	4.37 s	6.10 s
Dim (<150 lux)	Yaw Tilt (±15°)	14.40 s	13.15 s	13.77 s
Composite System Mean Response Time Across All Operational Scenarios				**6.00 s**

Given that the identity check is executed under very bright light, the execution lasts only 3.80 seconds. However, in the conditions of low light and yaw rotation, the execution time grows to 13.77 seconds due to localized searches of features in low-gradient pixels. In general, the system performance in all conditions was about 6.00 seconds, which is very important since the FOR was 0% meaning that each student passing the monitored checkpoint was identified properly.

B. Simultaneous Multi-Student Batch Verification In this part of the research, multi-student batches of 2, 4, and 6 students during the passage of the checkpoint were analyzed in terms of effectiveness. The multithreaded architecture allows the system to crop the bounding box and make the feature projection via ResNet at the same time. The results can be seen in Table-VI.

TABLE VI. Simultaneous Multi-Student Verification Throughput

Simultaneous Faces	Batch Detect (ms)	Parallel 128-D (ms)	KNN Match (ms)	Total Cycle (ms)	Batch Accuracy (%)
1 Student	42 ms	118 ms	4 ms	164 ms	100.0%
2 Students	58 ms	234 ms	7 ms	299 ms	100.0%
4 Students	84 ms	462 ms	14 ms	560 ms	98.5%
6 Students	112 ms	688 ms	22 ms	822 ms	97.2%

V.DISCUSSION & PRACTICAL INSTITUTIONAL IMPLEMENTATION

The empirical evidence shows that the Multi-Facet Institutional Attendance System (MFIAS) solves effectively the usability and security issues that have faced earlier attendance systems.

- The problem of QR Code Vulnerabilities has been solved. Unlike conventional QR-based systems necessitating manual student initiation with the aid of shared pictures, in MFIAS an autonomously acting Timetable Engine activates each session, thus eliminating the danger of proxy attendance.
- Resilience under Multi-Factor Variations. By synthesizing Dlib 68-landmark structural positioning with ResNet-34 metric embedding, MFIAS guarantees 99.29% recognition accuracy despite various disturbances introduced by the students daily, including glasses wearing, growing beard, and head movements.
- There are no false omissions even when MFIAS works continuously. Despite the fact that Haar Cascade detection may display some static frame misses if applied at intricate angles, the timely redundancy of continuous 30 fps video stream allows authenticating all legitimate students during a 6 second period with zero false omissions.
- Data Governance and Auditability: In order to prevent any manipulation of grades, writing access is limited to the administrators alone which ensures that unauthorized changes cannot be made. At the same time, rapid reporting of information across multiple vectors, including CSV, Excel, and PDF meets the regulatory requirements of the universities.
- Practical Deployment Considerations: When deploying MFIAS technology in large lecture halls hosting more than 200 students, using a single camera system may lead to delays during the five-minute peak hour traffic. In support of this view, the option to have a second camera installed alongside the current setup will enable the possibility to ensure faster data processing through having multiple access points.

VI.CONCLUSION & FUTURE SCOPE

The article outlines the architecture, implementation, and empirical assessment of the MFIAS, a self-reliant and non-contact facial recognition technology. The introduction of the Automated Department Timetable Engine has eliminated manual methods of QR scanning in the educational institutions environment. At the core of the system is biometrics technology, utilizing Haar Cascade face detection, Dlib 68 facial points alignment, ResNet-34 128-dimensional vector formation, and K-NN Distance weights classification. The use of this system provides an amazing 99.29% accuracy in video recognition, 6 seconds response time, and 0% omissions despite various lighting conditions, poses, and accessories.

Furthermore, the system tracks automated In/Out times with auditable administrative access via reports in various formats.

Future enhancements plan to expand the developments in three areas:

(i) lightweight infrared (IR) depth camera technology with eye blink CNN classifiers for hardware-level anti-spoofing;

(ii) (ii) installation of multi-camera edge nodes with TensorRT acceleration for reducing inference latency to under one second in big auditoriums; and (iii) incorporation of predictive analytics technology for the early identification of cases of chronic student absence.

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DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

Mrs. V. Devinandhini, M.E
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
devinandhini@psr.edu.in

Kannan.A
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
kannanandiappan695@gmail.com

Muthu Bala.R
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
balamuthu469@gmail.com

Santhosh.M
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
santhosh3122005@gmail.com

Akash.K
Department of AI&DS
P.S.R. Engineering College
Sivakasi, Tamil Nadu, India
akashakash98047@gmail.com

ABSTRACT

Conventional techniques for tracking student attendance in educational institutions typically rely on the old-fashioned manual attendance method, the use of physical sign-in sheets, or simple biometrics such as fingerprint scanning. The methods employed in traditional attendance monitoring are known to be laborious, disruptive to lessons, vulnerable to human fraud, and unhygienic in the case of a large group of students. New developments in contactless technologies have made it possible to use QR codes with web-based applications but the novel method still has its issues including the need to avoid queues, the chance of fraud detection, and the requirement for someone to point a QR code this person's phone.

The Multi-Facet Institutional Attendance System (MFIAS) is being introduced in this paper as a contactless attendance system that is dependent on schedule which works all problems present in conventional systems. The manual scanning of people during attendance is not applicable any more thanks to the Automatic Departmental Timetable Engine which is scheduling classes and monitors them according to it. Final identity verification is performed by means of the multi-faceted computer vision that considers many real world possible hindrances like dynamic lighting in classrooms, various head position angles (tilt ± 15 degrees), and many more things that may block the identification (like eyeglasses and beard). The process employs the usage of Haar Cascade framework for the image detection process, Dlib 68 points model for landmarks issues, Residual Neural Network RegNet-34 for defining facial features, and KNN (k=3) method.

Additionally, it contains asynchronous multistudent batch checking, precise date and time recording for arrival and departure as well as duration calculating. The attendance management software features strict authorization at the administrator level which makes it possible to update attendance record only by the appropriate people. Extensive experiments show that the video classification from still pictures as well as from the live and previously recorded video gives the accuracy of 99.29 percent, 0 percent false omission rate and 6 seconds average verification time despite the changes in videos. Also, the cloud attendance software is able to produce tamper-proof log files and convert them into CSV, Excel and PDF format files that helps assure the safety of attendance taking process through biometrics.

I. INTRODUCTION

Attendance records in schools are considered to be vital part of education management. Such records are known for helping in various internal evaluations, approval to conduct research in laboratories, and accreditation of examinations in educational establishments. Nonetheless despite numerous innovations in modern-day technologies participation is rather old-fashioned and is taken mainly in shape of roll call when people are asked to fill out sheets

with signatures or go through the process of using gadgets for collection of information about attendance such as fingerprint scanners. Despite the high popularity of the methods that are used in this area they cannot be called efficient because roll calls may take up to 10-18 percent of class time; paper sheets are still subject to the dangers of buddy signing, and gadgets require physical contact and cannot be used in all situations.

To overcome these challenges, creative approaches were introduced in the form of mobile applications and QR codes scanning systems. In the traditional QR code approach, a teacher generates a code which has a period after which it expires and a student scans it to show that he is present. However, research shows that QR codes did not eliminate cheating, they just made it different since students can taking pictures of a QR code on the screen and send it to their friends outside the university. Also, the necessity for students to locate and open the application, log in to the network using the application and point the camera at the screen introduces additional difficulties and 8 delays into the process of teaching.

Biometric facial recognition technology represents the primary way of performing academic identity verification without direct contact. After capturing human facial features, they cannot go missing or be altered; therefore, biometric facial recognition technology allows automatically keeping track of students in class just by watching them coming into the classroom. However, ready-to-use face recognition systems perform significantly worse in free unstructured educational environments. In ordinary classrooms students' faces may present a great variety of recognizable and unrecognizable features.

- **Changing Brightness:** Daylight differs from shining sunlight from the outside to dim light inside classrooms.

13

- **Tilting in Movement:** Students do not walk straight when passing through the doors of classrooms.

- **Facial Overlapping:** Common accessories of people, for example, eyeglasses with heavy glass, different types of beards and hats cover some important geometric reference points.

- **Delay in the Process of Ingress:** The usage of complex end-to-end implementation models causes significant costs resulting in the delay of data ingressing into the classroom.

14

This study develops an automated, contactless MultiFacet Facial Recognition Attendance System to eliminate the challenges associated with existing limitations without requiring students to scan QR codes. The system comprises five important parts:

1. The Timetable-Based Punctuality System allows direct access to the department's timetable which allows to automatically plan attendance sessions without relying on QR codes.

2. Advanced Biometric Algorithms allow creating sophisticated data processing systems involving Haar Cascade classifiers, Dlib's 68-point face modeling method, ResNet-34, and K-Nearest Neighbor algorithms.

3. The usage of Simultaneous Student Verification allows the acquisition of information from the video stream and recording of the concurrent presence of learners.

4. Dynamic Audit of Entry and Exit Times enables registration of the time of arrival and departure of students from the classroom, creating the complete record of their presence in class.

5. Admin Security prevents anyone else from performing writing and reporting operations within the database.

The rest of the paper is organized into sections. The following section describes the previously conducted research and the methods which form the basis for the technique described in this paper. This section is anything but a general outline of the mathematical approaches and methods utilized in the realization of the project. The section IV gives the evaluation of the technique and includes the practical assessment of the technology used. The next section V contains the evaluation of the technical part of the system as well as organizational issues. Finally, section VI gives the conclusions and recommendations for the future research.

II. RELATED WORK & RESEARCH FOUNDATIONS

Over the last 20 years, automated attendance systems have come a long way. In the early days, these systems made use of the Viola-Jones object detection algorithm together with either Local Binary Pattern Histograms (LBPH) or Principal Component Analysis (PCA) to overcome the challenges related to capturing localized facial features. While these systems do not require heavy computational resources and allow the implementation of attendance systems on microcontrollers, they fail in extreme lighting situations, slight head movements and obscured faces.

Automated attendance systems have made great progress in the last 20 years. At first, this technology was based on Viola-Jones object detection algorithm combined with Local Binary Pattern Histograms (LBPH) and Principal Component Analysis (PCA). Hence, the algorithms did not need much power and, therefore, were usable even on microcontrollers, but they still did not manage to produce reliable results in difficult lighting conditions, in particular, when head movements or obscured faces were present.

The emergence of deep convolutional neural networks (CNNs) enabled researchers to switch to high-

dimensional representation methods of analysis. Pattnaik and Mohanty [7] conducted an evaluation of cloud-based computer vision services such as Google Cloud Vision or Amazon Rekognition that proved to have high accuracy in picture analysis. When switching to real-time video transmission the disadvantages of such methods became obvious: video-steaming creates heavy traffic load and also has untrusted latency. Moreover, in the case, where privacy is an issue, it requires stable connection to the Internet of the university's camera. According to the study performed by Lateef and Kamil, it became possible to track many people's faces in one smart classroom with the help of different YOLOv7 models.

Nevertheless, the fact that depth detection devices must operate at real-time frame rates necessitates the need for expensive GPU technology, making these products impractical for use in schools, as well. So as to cope with a certain kind of blocking of the face, Kamil and his coauthors came up with a system for online authentication that is capable of detecting individuals that have protective masks as required by the condition of the pandemic, achieving an efficiency of 81.8% in identification of such individuals. Nevertheless, this system is not good for solving the case of complicated distortions and is aimed only at solving the problem of identifying the lower part of the face with the nose covered with a mask. In addition, Afiyanto and his co-authors substantiated the effect of variation of illumination on the metric Euclidean distance which is reduced by 34% within the classrooms that are constantly illuminated by 150 lux and less.

The research by Jack and others into the field of multidimensional analysis of attendance management has resulted in a major breakthrough that solves many existing problems. According to the claims of Jack et al., the decoupled hybrid solution composed of Haar Cascade detection, Dlib landmark alignment, ResNet-34 128dimensional vector encoding and distance-weighted KNN classifier shows 99.29% recognition rate in video streams and 6.00 seconds average verification time, while the other deep learning models are encountered with the obstacles caused by the shortcomings of the inference process.

TABLE I. Comparative Summary of Automated Attendance Literature

Reference	Core Methodology	Primary Advantage	Identified Limitations & Gaps
Bairagi et al. (2021)	Haar + LBPH	Minimal cost; fast writes.	Fails under off-pose & lighting shifts.
Trivedi et al. (2022)	OpenCV + Eigenfaces	Automates SQL logging.	Rigid alignment; zero occlusion tolerance.
Kamil et al. (2023)	SVM + Mask CNN	Masked face verification.	Isolated occlusions only; no pose handling.
Lateef & Kamil (2023)	YOLOv7 Deep Model	Continuous multi-face tracking.	Heavy GPU load; high latency at edge nodes.
Afiyanto et al. (2025)	Dynamic Filter + CNN	Illumination normalization.	High overhead; fails on composite angles.
Jack et al. (2026)	Haar + ResNet-128D + KNN	Multifacet robustness (99.29%).	6.00s latency; lacks timetable automation.
Proposed MFIAS	**Timetable + ResNet + KNN**	**Zero QR; batching; In/Out logs.**	**Needs calibrated entry camera placement.**

III. PROPOSED SYSTEM ARCHITECTURE & METHODOLOGY

The Multi-Facet Institutional Attendance System or MFIAS has been developed with six modules that are interrelated to each other in a unique way. The modules are; (i) Student Biometric Enrollment Layer; (ii) Departmental Timetable Scheduling Engine; (iii) Multi-Students Computer Vision Ingestion Layer; (iv) Multi-Facet Deep
Feature Embedding and Classification Engine; (v) In/Out Temporal Tracking and State Machine; and (vi) RoleBased Admin Governance and Multi-Format Export Module.

A.Student Enrollment & Biometric Blueprint Synthesis

Enrollment can be viewed as an administrative procedure that takes place before the operation of classes begins. Every student accepted by the institution is registered with his/her demographic details such as Roll Number, Name, Department, Year, and Academic Section. Enrollment is carried out in front of the camera, which captures 30 different images of the student's face.Frontal gaze under

ambient and supplemented lighting. • Yaw motion involves tilting forward and backward (by 15°).

• The study would consider the impact of wearing and not wearing glasses along with different types of facial hai

f.

The images indexed are processed through bilinear interpolation to achieve a regulated affine alignment of 400 × 400 pixels in size. All images are then converted to singlechannel 8-bit grayscale format, thus effectively capturing the texture and geometry while disregarding color variations. Then, facial landmark points are extracted, converted to 128-dimensional embedding vectors by the ResNet-34 deep learning backbone, and stored in an indexed binary file through Pickle available in the Python environment. This way, the need for deep learning training in real time has been eliminated due to the pre-computed embeddings.

B. Automated Departmental Timetable Engine

A significant advancement brought about by MFIAS is the removal of the reliance on QR codes and mobile scans inside the elements. The attendance of students should adhere to the established formal schedule of classes. The Timetable Engine functions as an independent daemon thread interacting with the university database. The engine relies on the master schema of relationships:

```
TIMETABLE_SLOT = {
"slot_id": UUID,
"department": "AI&DS",
"semester": 6,
"section": "A",
"day_of_week": "MONDAY",
"start_time": "09:00:00",
"end_time": "09:50:00",
"grace_period_mins": 10,
"course_code": "AD8601",
"course_title": "Computer Vision & Deep Learning",
"assigned_faculty_id": "FAC_9519_01"
}
```

The hardware clock is compared to the timetable index all the time:

1. **Pre-Session Trigger:** In advance of the time of the session by 2 minutes, the system starts the camera pipeline, sets buffer streams, and saves the biometric vectors of students who are enrolled in the class from the respective department and semester.

2. **Effective Presence Window:** During the period from initial time to final time, video feeds are analyzed to assess student presence. Students verified during grace minutes are marked as Present; verifications occurring after the grace time are designated Late.

3. **Completion of Session:** After end time, the camera feeds are automatically disabled. Any student participating in the group who was absent during that period is logged as Absent. The final attendance log is saved.

C. Multiperson Face Recognition and Alignment

For the whole time of the session, the camera output is recorded at a rate of 30 frames per second (fps). Unlike the single portal cases, it should be noted that in reality the students do not walk alone. The platform uses OpenCV pretrained frontal Haar Cascade classifier for the processing of the video stream. The system uses integral image representation to assess rectangular Haar-like features through a multi-scale sliding window: $ii(x, y) = \sum x' \leq x, y' \leq y\, i(x', y')$ **(1)**

In this notation, ii(x,y) refers to the cumulative value associated with facial recognition, while i(x',y') indicates the pixel intensity. The methodology is based on a series of AdaBoost classifiers eliminating undesirable image backgrounds in the earlier stages of analysis. All the bounding boxes Bk = [xk, yk, wk, hk] are acquired at once.

To achieve the required consistency despite head inclinations up to ±15°, the bounding box coordinates are aligned to the facial features of the person. A Dlib ensemble of regressor decision trees is used to determine the location of 68 key facial features including chin, eyebrows, nose, eyes, and mouth. The transformation between the inner corners of the eyes (features 37 and 46) and the tip of the nose (feature 31) is calculated and is used to transform the face to a frontal position.

D.Deep Feature Extraction & ResNet-34 128-D Metric Encoding

Affinities of normalized facial regions are learned using the Residual Neural Network (ResNet-34) architecture. Instead of performing identity classification like a softmax cross-entropy classification problem, our idea is to use the network purely as a feature extractor and convert facial landmarks based on geometry into 128-dimensional Euclidean space:

$$f: I \rightarrow \Re^{128} \quad ext\{such\ that\} \quad \|f(I)\|_2 = 1 \qquad (2)$$

The ResNet-34 architecture has shortcut identity connections over 34 convolutional layers, allowing us to train deep networks without the problem of gradient vanishing. The weights are trained with the aid of the Triplet Loss method. The triplet Loss function makes it necessary for similar facial representations (anchor f(xia) and positive case f(xip) to be grouped close to each other and meaning that different facial representations (negative case f(xin)) must be separated from each other with a predetermined distance α:

$$L = \sum_i^N max(0, ||f(x_i^a) - f(x_i^p)||_2^2 - ||f(x_i^a) - f(x_i^n)||_2^2 + \alpha) \quad (3)$$

In our case, the value of α is equal to 0.2. By using this approach, the variations of the similar person facial appearance caused by eyeglasses, beard, or bad lighting create a minimal shift in the vector that is still located within the same identity area but separate from the identities of other students because of the high distance in Euclidean space.

E. Distance-Weighted K-Nearest Neighbors (KNN) & Unknown Rejection

For live identity assignment, the normalized 128dimensional vector x undergoes classification against enrollment symbols recorded in the database using the distance-weighted K-Nearest Neighbors algorithm with k = 3. The Euclidean distance calculated between this test vector and the training vector xj can be written as: $d(x, x_j) = \sqrt{(\sum_{m=1128} (x_m - x_{j,m})_2)}$

In order to minimize the impact of outliers caused by changing lighting conditions, the voting weights wj are assigned inversely proportional to the square of the Euclidean distance to the enrolled individual:

$$w_j = 1 / (d(x, x_j)^2 + \varepsilon)$$

Where ε = 10-5 avoids division by zero. The predicted identity y is found through the majority voting mechanism:

$$y = argmax_c \sum_{j=1}^k I(y_j = c) \cdot w_j$$

The Unknown Person Protocol: In order to avoid the cases of false positive identification, strict limitations regarding the Euclidean distance threshold are established.

```
IF min(d(x, x_j)) <= 0.60:
    Identity = y (Valid Student
Verified)
ELSE:
    Identity = "UNKNOWN / INVALID
STUDENT"
    Trigger Alert Flag & Log
Unregistered Face Capture
```

Temporal In/Out State Machine & Session Tracking

Contrary to single-instance check-ins, the requirement of educational accountability entails checking the time spent by each student inside the classroom. The MFIAS database captures 2 separate time stamps for individual students for each session.

1.In-Time (Entry Timestamp): This time stamp is generated as soon as a student is detected in the ongoing session for the first time:

entry_timestamp = CURRENT_TIMESTAMP.

2. Out-Time (Exit Timestamp): The last time stamp is updated every time a student is detected in the last 10 minutes of the session or when he/she leaves the class: exit_timestamp = CURRENT_TIMESTAMP.
3. Duration Calculation: Once the session is over, the duration is calculated as follows: duration_minutes = (exit_timestamp - entry_timestamp)/60 4. Attendance Status:

○ If entry_timestamp ≤ (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as PRESENT.
○ If entry_timestamp > (start_time + grace_period) and duration_minutes ≥ 0.75 × session length, then it is recorded as LATE.
○ If it is neither detected nor duration_minutes < 0.50 × session length it is recorded as ABSENT.

G. Role-Based Administrative Governance & MultiFormat Reports

It is important for the security integrity to ensure that neither faculty members nor students can modify attendance records. MFIAS also uses JWT authenticationbased RBAC.

- Student/Public Role – Read-only access to personal attendance percentages.
- Faculty role – Read-only access to session summaries and attendance records in the classroom.
- Admin role – Only one who has the ability to reset attendance flags (such as being on duty). Administrative changes are recorded through explanation codes whereby the ID of the administrator is also logged in the system.

For archiving purposes, the website has embedded report generators that automatically generate reports in three different formats.

- CSV – Spreadsheet format with raw data.
- XLSX – Tables with colors coded status stamps and statistics of attendance.
- PDF – Printed reports with signatures.

IV. Experimental Setup, Results & Performance Evaluation

A.Experimental Environment & Dataset Curation

Real-life testing in an academic environment was simulated in a laboratory setting. Technology used in the computational testbed included an Intel Core i7 CPU, 16GB DDR4 RAM, and NVIDIA GeForce RTX 3060 GPU (for the ResNet embedding) and the processing of the CPU was employed for answering the question of whether edge computing is feasible.

The experimental data were collected utilizing the method presented by Jack et al. (2026). The images were created with various experimental conditions in mind (illumination (bright vs dim), head position (facing forward, tilted to the right or left at a 15-degree angle), clothing (with or without glasses and beard). The evaluation included more than 3000 still images, over 5173 video frames, and real-life tests of continuous student entry.

B. Detection Stage Evaluation: Failure to Detect (FTD) Dynamics

System evaluation requires decoupling face detection from classification. Table II outlines the detection efficacy of the pre-trained Haar Cascade classifier across static frames and continuous video.

TABLE II. Haar Cascade Face Detection Performance Metrics

Evaluation Modality	Total Samples	Detected Samples	Detection Rate (%)	FTD Rate (%)
Static Image Dataset	3,000	1,750	58.33%	41.67%
Continuous Video Frames	5,173	4,059	78.46%	21.54%

It was noted that static photographs had an FTD of 41.67%, partly as a result of acute head positions and dim areas of low contrast that did exceed gradient limits set by Haar-like filters. In contrast, during regular video transmission (30 fps), the FTD dropped to 21.54%. Importantly, an FTD event does not result in a false acceptance; it simply rejects a frame that is unsuitable for further processing. Due to the temporally redundant nature of live broadcasts, a passerby walking through a doorway will yield anywhere from 60 to 120 frames for accurate detection as the student adopts a natural position.

C. Evaluation of Multilevel Recognition Performance

To evaluate the quality of classification, 4,059 frames correctly localized over video recordings were examined. The particulars of precision, recall, and F1score for each student group are presented in Table III.

TABLE III. Recognition Performance Metrics on Video Frames

Student Identifier	Frames	Precision	Recall	F1-Score	Accuracy (%)
Student Cohort 1	1,215	0.9959	0.9934	0.9946	99.41%
Student Cohort 2	492	0.9979	0.9593	0.9782	98.78%
Student Cohort 3	2,352	0.9903	0.9996	0.9949	99.36%
Cumulative Average	**4,059**	**0.9947**	**0.9841**	**0.9892**	**99.29%**

The system demonstrated that the 128-dimensional ResNet-34 embeddings yield good separation between different classes while remaining close within classes regardless of the type of accessory or slight motion blur, since the overall recognition accuracy achieved by the system was 99.29%.

A. Classifier Testing: Comparison of KNN and Other Machine Learning Models To confirm the appropriateness of using distance-weighted KNN as the architecture, the same 128-dimensional embedding vectors were used to assess the classification accuracy of a Linear Support Vector Machine (SVM) and a Shallow Multi-Layer Perceptron (MLP with a hidden layer of 64 neurons running in Adam optimization mode).

TABLE IV. Comparative Benchmarking of Lightweight Classifiers

Classifier Head	Static Acc (%)	Video Acc (%)	Precision	Recall	F1-Score	Retraining Need
Linear SVM	99.94%	99.12%	0.9937	0.9749	0.9842	Full Retrain
Shallow MLP (64)	99.66%	99.25%	0.9937	0.9812	0.9931	Iterative BP
Weighted KNN (Ours)	**99.94%**	**99.29%**	**0.9947**	**0.9841**	**0.9892**	**Instant Insert**

Despite their accuracy scores being only slightly below that of KNN (99.12% and 99.25% for SVM and MLP respectively), in practice, the algorithms of both of these models should be re-tuned across the entire student base whenever a new student appears. In this regard, the distance-weighted KNN is a non-parametric algorithm that allows for an unrestricted addition of new students into the search space without affecting the speed of inference. *Response Latency*

& Multi-Facet Operational Robustness The real-time working capacity of the system was analyzed in different conditions to determine its response speed and False Omission Rates (FOR). The data is provided in Table V.

A.TABLE V. Average Response Time (Seconds) Under Multi-Facet Variance

Illumination Setting	Head Orientation	With Glasses / Beard	Without Glasses / Beard	Avg Latency (s)
Bright (>450 lux)	Frontal Straight (0°)	3.79 s	3.81 s	3.80 s
Bright (>450 lux)	Yaw Tilt (±15°)	4.37 s	4.11 s	4.24 s
Dim (<150 lux)	Frontal Straight (0°)	7.84 s	4.37 s	6.10 s
Dim (<150 lux)	Yaw Tilt (±15°)	14.40 s	13.15 s	13.77 s
Composite System Mean Response Time Across All Operational Scenarios				**6.00 s**

Given that the identity check is executed under very bright light, the execution lasts only 3.80 seconds. However, in the conditions of low light and yaw rotation, the execution time grows to 13.77 seconds due to localized searches of features in low-gradient pixels. In general, the system performance in all conditions was about 6.00 seconds, which is very important since the FOR was 0% meaning that each student passing the monitored checkpoint was identified properly.

B. Simultaneous Multi-Student Batch Verification In this part of the research, multi-student batches of 2, 4, and 6 students during the passage of the checkpoint were analyzed in terms of effectiveness. The multithreaded architecture allows the system to crop the bounding box and make the feature projection via ResNet at the same time. The results can be seen in Table-VI.

TABLE VI. Simultaneous Multi-Student Verification Throughput

Simultaneous Faces	Batch Detect (ms)	Parallel 128-D (ms)	KNN Match (ms)	Total Cycle (ms)	Batch Accuracy (%)
1 Student	42 ms	118 ms	4 ms	164 ms	100.0%
2 Students	58 ms	234 ms	7 ms	299 ms	100.0%
4 Students	84 ms	462 ms	14 ms	560 ms	98.5%
6 Students	112 ms	688 ms	22 ms	822 ms	97.2%

V.DISCUSSION & PRACTICAL INSTITUTIONAL IMPLEMENTATION

The empirical evidence shows that the Multi-Facet Institutional Attendance System (MFIAS) solves effectively the usability and security issues that have faced earlier attendance systems.

- The problem of QR Code Vulnerabilities has been solved. Unlike conventional QR-based systems necessitating manual student initiation with the aid of shared pictures, in MFIAS an autonomously acting Timetable Engine activates each session, thus eliminating the danger of proxy attendance.
- Resilience under Multi-Factor Variations. By synthesizing Dlib 68-landmark structural positioning with ResNet-34 metric embedding, MFIAS guarantees 99.29% recognition accuracy despite various disturbances introduced by the students daily, including glasses wearing, growing beard, and head movements.
- There are no false omissions even when MFIAS works continuously. Despite the fact that Haar Cascade detection may display some static frame misses if applied at intricate angles, the timely redundancy of continuous 30 fps video stream allows authenticating all legitimate students during a 6 second period with zero false omissions.
- Data Governance and Auditability: In order to prevent any manipulation of grades, writing access is limited to the administrators alone which ensures that unauthorized changes cannot be made. At the same time, rapid reporting of information across multiple vectors, including CSV, Excel, and PDF meets the regulatory requirements of the universities.
- Practical Deployment Considerations: When deploying MFIAS technology in large lecture halls hosting more than 200 students, using a single camera system may lead to delays during the five-minute peak hour traffic. In support of this view, the option to have a second camera

installed alongside the current setup will enable the possibility to ensure faster data processing through having multiple access points.

VI.CONCLUSION & FUTURE SCOPE

The article outlines the architecture, implementation, and empirical assessment of the MFIAS, a self-reliant and non-contact facial recognition technology. The introduction of the Automated Department Timetable Engine has eliminated manual methods of QR scanning in the educational institutions environment. At the core of the system is biometrics technology, utilizing Haar Cascade face detection, Dlib 68 facial points alignment, ResNet-34 128-dimensional vector formation, and K-NN Distance weights classification. The use of this system provides an amazing 99.29% accuracy in video recognition, 6 seconds response time, and 0% omissions despite various lighting conditions, poses, and accessories.

Furthermore, the system tracks automated In/Out times with auditable administrative access via reports in various formats.

Future enhancements plan to expand the developments in three areas:

(i) lightweight infrared (IR) depth camera technology with eye blink CNN classifiers for hardware-level anti-spoofing;

(ii) (ii) installation of multi-camera edge nodes with TensorRT acceleration for reducing inference latency to under one second in big auditoriums; and (iii) incorporation of predictive analytics technology for the early identification of cases of chronic student absence.

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Submission Summary

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Paper Title
DEVELOPMENT OF A MULTI-FACET FACIAL RECOGNITION MODEL FOR INSTITUTIONAL ATTENDANCE ACTIVITIES

Abstract
Conventional student attendance methods such as manual registers, physical sign-in sheets, fingerprint scanning, and QR-based systems are time-consuming, disruptive, vulnerable to proxy attendance, and difficult to manage for large classrooms. This paper presents the Multi-Facet Institutional Attendance System (MFIAS), a contactless, schedule-driven facial recognition system designed for automated and reliable attendance management. The system integrates an Automatic Departmental Timetable Engine to schedule classes and monitor attendance sessions without manual scanning. For identity verification, a multi-faceted computer vision pipeline handles practical variations such as dynamic classroom lighting, head pose variations up to ±15 degrees, eyeglasses, and beard-related appearance changes. The proposed pipeline uses Haar Cascade for face detection, the Dlib 68-point model for facial landmark alignment, RegNet-34 for facial feature extraction, and K-Nearest Neighbors (KNN, k=3) for identity classification. The system supports asynchronous batch verification of multiple students, accurate recording of arrival and departure times, duration calculation, and role-based authorization. Experimental evaluation using still images, live video, and previously recorded video demonstrates a recognition accuracy of 99.29%, a false omission rate of 0%, and an average verification time of 6 seconds. The system also generates tamper-resistant attendance logs and supports CSV, Excel, and PDF exports. The results demonstrate that MFIAS provides an efficient, contactless, and robust approach for institutional attendance automation.

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Authors
Kannan A (P.S.R Engineering College) <kannanandiappan695@gmail.com>
Devinandhini V (P.S.R Engineering College) <devinandhini@psr.edu.in>
Muthu Bala R (P.S.R Engineering College) <balamuthu469@gmail.com>
Santhosh M (P.S.R Engineering College) <santhosh3122005@gmail.com>

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